

Karl Friston ~ Active Inference Insights 001 ~ Free Energy, Time, Consciousness

Insights | Active Inference Insights 001 ~ Free Energy, Time, Consciousness | 2:48:25

Excellent. And we're away. Okay. So welcome everybody to the first episode of Active Inference Insights brought to you by the Active Inference Institute. I am your host, Darius Parvizzi-Wayne, and today I have the privilege of speaking to Professor Carl Friston, one of the most revolutionary thinkers in the fields of neuroscience, cognitive science, theoretical biology, and dare I say physics and philosophy. A leading authority in neuroimaging, he is the inventor of statistical parametric modelling and voxel-based morphometry. I hope I pronounced that correctly. More recently, he has posited the free energy principle, which is technically a normative account of self-organisation in terms of optimal Bayesian design, or in other words, a description of how things continue to be things. As such, his groundbreaking ideas hold profound implications not only for biology and psychology, but also the very nature of reality itself. Ladies and gentlemen, it is my honour to introduce Professor Carl Friston. Professor Friston, thank you so much for joining us.

Well, thank you. Thank you for that lovely introduction. But what you can say is I just invented lots of three-letter acronyms. That's much simpler.

So you have. And I sense a lot of three-letter acronyms have also spawned from a lot of your work.

Excellent. Well, this podcast, this is the first episode, as I was saying, sort of off-camera, this is very exciting for me,

hopefully very exciting for everyone tuning in. The reason why it's come about is that we wanted, the Active Inference Institute wanted a way to introduce the ideas of active inference and the free energy principle,

perhaps to an audience who were not PhD holders, but had a kind of latent interest in psychology, neuroscience, maths, biology, whatever it is, and wanted to explore these ideas a little bit further.

So it's a non-specialised introduction to the ideas of active inference and the free energy principle.

And as the author, inventor, discoverer of the free energy principle, we can have an interesting philosophical conversation about what that is.

It's an absolute delight to have you.

So I guess my first question that I need to ask for everyone listening is, could you give us a very brief, or not very brief, but a very accessible definition of the free energy principle, what free energy is, and what this principle says about reality and ourselves?

Yes, I'm not very good at giving brief answers, but I'll do my best.

We have a while, so go ahead.

So just to put things in context, the free energy principle is a principle as read by a physicist or engineer.

That means it's a method, much like Hamilton's principle of least action, that you can apply.

So one might ask, what are the most important applications of the free energy principle?

I guess that would be active inference.

So I will answer your question, and I will say what the free energy principle is,

but that may have to be qualified by looking specifically at the application, which is active inference, and what that means teleologically, heuristically, what it brings to the table when you do apply the free energy principle. But the free energy principle itself is just a description of things that persist or exist, and it rests upon a careful specification of what you mean by a thing. It starts with exactly the same assumptions that all of physics starts with, basically a random dynamical system. Some people describe that in terms of stochastic differential equations. If you're not familiar with that kind of description, it's just saying that you can describe the universe in terms of the way that states change as a function of themselves. So it's a very generic description of any given universe that just commits to the notion that there is a state of the universe, and it could be very high dimensional, and that it has some dynamics. So that's where the free energy principle starts. But the special thing about the free energy principle that distinguishes it from other formulations in physics, for example, quantum thermodynamics, statistical and classical mechanics, all of which can be, if you like, derived from this basic description of the world as a random dynamical system. The thing that separates the free energy principle, or gives it its particular flavor, is a careful distinction between the states of something and everything else, and the way that we do that is by inducing a separation, technically a partition of all possible states into three kinds, the states of something that are internal to the thing, everything else external to the thing, and then crucially, a set of states that separate the internal from the external states that allow and couple the inside to the outside, and that's referred to as the blanket states, or a Markov blanket or boundary. So a lot of this rests upon this notion of a boundary that, on the one hand, separates the inside from the outside of something, and by a thing, I mean anything from a small particle to a priest or a person, anything that can be individuated and possesses characteristic states, constitutes some thing. And with tongue-in-cheek,

you therefore have a theory of every space thing,
not everything,
but everything.

So this Markov boundary or Markov blanket is a sort of crucial device
in the sense that it allows you to individuate something from everything else,
but at the same time,
it couples the internal to the external states.

So we're now talking about a special kind of physics,
probably a more ubiquitous kind of physics,
which is the physics of open systems,
because the inside is now vicariously open to,
in exchange with,
the outside across the Markov blanket.

And just for interest,
because this will be relevant later on for active inference,
we usually divide the blanket states into active and sensory states
that can be regarded very simply as on the inputs and the outputs.

Technically,
the inputs of the sensory states are defined in the sense that the internal states
don't influence the sensory states.

And conversely,
the active states are defined such that the external states don't influence the active states.

So,
but more simply,
what we're saying is that something can be defined in terms of the influences upon it
and the influences that it exerts through the blanket states on the outside world.

So,
just given that assumption that entails existence in the sense that to define this Markov blanket,
you have to,

you need for the system to have some characteristic states,
technically,

an attracting set of states that you'd find the thing in,
just by assuming the existence of this attracting set of this random dynamical system
that possesses this distinction between the inside and the outside,
everything else follows.

and everything else,

simply put,

is you can read the internal and active states,

sometimes referred to as the autonomous states of a particle or a priest or a person,
as either,
if you're a physicist,
conforming to a particular principle of least action,
and that principle of least action is that it minimizes the path integral of the variation of free energy.
If you want to think of this more intuitively,
because that free energy also stands in for,
or is an approximation to,
a bound upon model evidence,
you can also say that it must be the case
that the autonomous states of anything
will look as if they are trying to maximize the evidence
for what we would now consider,
the internal states to be a model of what's going on on the outside.

So,
some people refer to that as self-evidencing,
so if I just repeat that from the physicist's point of view,
given the assumption that you've got this random dynamical system
with an attracting set,
that can,
that has this partition that allows you to individuate something within this system,
it has to be the case
that the dynamics of the autonomous states of that system
perform a gradient flow on a quantity called variation free energy.

That's it.

Is that interesting?

Well, it becomes interesting
if you realize that this variation free energy
has a particular meaning
and a particular interpretation
that gives you a teleology for this kind of dynamic,
and that teleology is often referred to in terms of Bayesian mechanics.

Why?

Well, because the free energy,
the variation of free energy,
is the same quantity that measures the evidence for a generative model of the causes of some data,
if you're a statistician.
And in this instance,

the data that we're talking about are the impressions of the outside state on the Markov blanket.

So what you're now in a position,

well, you're now in a position,

you are now licensed to talk about anything that exists in some simple elemental sense,

performing inference and possessing this or evincing this kind of Bayesian mechanics.

And in philosophy,

people like Jacob Howie nicely summarizes this as self-evidencing.

So you can imagine that it looks as if anything that exists is basically moving in a way,

sensing and acting upon its world in a way that's garnering evidence for its own existence.

So this is, you know, if you like, a slight cheat in the sense of, you know,

what we're doing is describing the dynamics of something that exists.

And then we're saying, but it now looks as if, in virtue of its existence,

it is now, it can be described, teleologically,

as acting to gather evidence for its own existence, its own structure.

And that structure entails a generative model of what's going on on the outside.

So how does that help?

Well, it doesn't really help in any fundamental way other than it's very pretty.

And it's nice to talk about Bayesian mechanics as being a sort of complement

or another kind of physics that comes from the same stable as quantum mechanics

and system mechanics and, I repeat, classical or Lagrangian mechanics.

However, there is a nice application at hand once you realise that this,

once you interpret the variation of free energy in terms of technically the logarithm of some model evidence.

What that means is that you can now build things.

You can build things and you can simulate those things.

Why? Well, because you know that their dynamics must be conforming to this principle of least action.

Another way of thinking about this is that it does a gradient descent or hill climbing on a particular functional,

which is this variation of free energy or the negative variation of free energy,

which is the log evidence.

So because this log evidence is a function of a model that just is a description of the characteristic states things occupy or things are found in,

you can now write down the generative model, which is just a probability distribution over the characteristic states

or defining the characteristic states you want your thing to be attracted to.

This is the attracting set.

And once you've written down your generative model, you can then work out the gradients of the free energy functional of that generative model.

And then you can simply integrate or mathematically solve for the dynamics, the flow, the motion, the movement of this thing.

So I know this isn't terribly accessible, but it's the way I think of these things that you are now in a position, instead of just looking at something and saying, oh, I wonder what generative model would be fit for purpose to describe this particular thing.

You can now turn that on its head and say, OK, I want this kind of thing.

And I'm going to describe and define this kind of thing with a particular generative model.

I just have to write down the probability distribution over the causes and consequences from the point of view of the thing in question,

namely the external states and its impressions upon the Markov blanket.

And once I've written that down, I can now simulate the autonomous states.

I can now create a little autonomous particle that behaves in exactly the same way as something that existed that had that kind of generative model.

So it's a very practical, if you like, it's a principle that can be applied in a very practical way to simulate things.

And if you can then simulate things, you can start to build things, you can start to do computational neuroscience,

you can start to sort of naturalise cognitive neuroscience and possibly even sort of psychology in terms of the physics that is inherent in this Bayesian mechanics

that comes from the free energy principle.

You can also start to ask questions about the beliefs of another person.

Why? Well, part of this generative model or the generative model can always be sort of split into two things, sort of priors before you see some data and the likelihood, the likelihood of that data given the causes of that data.

This would be the external states.

So you have prior belief over the causes, the external states, and you have a likelihood that maps the causes to the consequences,

which are your observations, which are your sensations.

And therefore, you can now fiddle with, and say, a synthetic subject or person who is being simulated in the computer

until their behaviour, their actions, possibly their decisions, matches that of any given cohort or any given subject or animal.

And then having adjusted the priors part of the generative model and possibly the likelihood, you've now basically created a digital twin that emulates the same kind of belief updating, the same kind of Bayesian mechanics that you can now infer is possessed by the thing in question.

So this is a sort of quite a high-end application, which is one of the motivations for developing the free energy principle

and active influences and application that enables you to essentially phenotype another thing, another person, for example,

in terms of this belief updating, in terms of this Bayesian mechanics.

And that can be quite useful in contexts when you want to understand how people work and particularly how their belief updating works.

This becomes particularly interesting in the context of computational psychiatry, where you want to be able to explain people's behaviour in terms of what do they believe about the world out there?

How are they sense-making? How are they assimilating evidence for their world model, which you can read as the generative model?

And under what priors are they doing this sort of Bayesian assimilation or evidence accumulation that underwrites their decisions?

So that's one particular application. There will be other applications, you know, if you want to make intelligent artifacts

or you want to understand the mechanisms of belief updating at different temporal scales, belief sharing between two things that are in communication or transacting across there, a shared Markov blanket or indeed looking at a shared world.

And then one can start to get into the world of distributed cognition and federated inference. There are all sorts of things you can do once you start to apply active inference.

Yeah, that was fantastic. I mean, we have a lot to get our teeth into.

And probably some concepts there that are new to our audience.

So it might be worth sort of backtracking certain things and trying to find a kind of, yeah, let's sort of dissect some of these concepts together.

The first thing I want to speak about is this notion that the internal states in some sense embody the kind of desirable model given their attractor state, the desirable model of the world.

I think one of these, one of the questions that one has when one enters into the world of active inference is whether an organism is a model or whether an organism has a model. And here we sort of have a backtrack to cybernetics and the good regulator theorem that all good regulatory systems have a model of their environment, if we think about thermostats or anything alike. So maybe it's worth unpacking that because I think a lot of people might have this intuitive sense that there's a homunculus in the brain, which has a model of the world and it's very internalist. But maybe from a more physics perspective, we can adopt a more externalist position where there isn't this little thing, which has a sort of representative picture of the world, but actually is just enacting some model of itself, which in many ways is just some version of its external dynamics.

So let's start there because I think that will feed nicely into maybe a conversation about internal and external dynamics and exactly what that looks like in the free energy principle.

Yeah, that's a great question. And you're some sort of deep issues unearthed by that. And it's nice you've introduced cybernetics, the notion of the good regulator theorems. You know, in one, on one view, I think you can look at the free energy principle as just a 21st century version of that kind of cybernetic thinking. There are other things that the free energy principle inherits from, you know, James's maximum principle, lots of ideas in psychology, perceptions, hypothesis testing. There are lots of different roads that converge upon this formulation that is on offer in terms of the free energy principle. But that particular question is really interesting. I have to say that

I usually try to elude that question by saying that the internal states entail a generative model. However, if one was pressed upon it, then I would say very clearly that the internal states, the thing is a generative model. It's not, if you like, you know, a picture that you painted in terms of that homunculus looking at it, sort of, you know, its virtual reality inside its head and trying to predict what's going on. It is instantiated, it is realized in its physics, in its existence, in its dynamics. The reason I use the word entailed is that the generative model is technically just a probability distribution.

So you ask yourself, well, does a probability distribution exist in some physical sense? And the answer is, well, no, a probability distribution is what it is. It's a mathematical description. However, that probability distribution will have sufficient statistics that could exist. So what I'm saying is that, for example, let's take a Gaussian distribution. You know, I can express mathematically, I can write down the notation for your Gaussian, a bell-shaped distribution over some variable that goes from plus to minus infinity. Or I can say, I can characterize or specify this Gaussian distribution in terms of its mean and its variance. And those are two numbers that are not random variables. They are sufficient statistics. And it may be, and in fact, it is the case that the free energy principle associates the physical internal states with the sufficient statistics of these Bayesian beliefs.

And there's a lovely story, you know, in mathematics that would take you down the path of information geometries and statistical manifolds and belief updatings as movements in these geometries. And, you know, trying to understand the metrics in these spaces and how that relates to uncertainty and precision. Also, that's a lovely story. We don't need to really pursue that at the moment. But the key, the point being made here, the generative model is not physically instantiated other than in terms of some sufficient statistics in any setting. However, in the free energy principle, the generative model is in order of itself never actually instantiated in terms or realised in terms of its sufficient system. And that's because of something that we were talking about before, which is the dynamics. So the dynamics is a gradient flow, which means that all we need are the gradients of the free energy. So we don't need the generative model. All we need is the gradients of this free energy function or technically a function or a function of a function of the generative model. So if I was building an artifact, I could certainly prescribe the right dynamics by writing down a generative model. And then I could evaluate the gradients of that free energy function of the generative model. And then those gradients would actually drive the dynamics. I could simulate self-organisation, I could simulate active inference, I could simulate belief sharing, whatever I wanted to do. But in that simulation, all I really needed were the gradients of the free energy.

And we only need the generative model because the free energy is a function of the generative model. So in that sense, the generative model is really a theological description that you would bring to the table to understand your sense-making of another thing. The thing in and of itself just needs the free energy gradients. It just needs to self-evidence by moving in the direction of maximising the, on average, the log evidence for some abstraction, which is a generative model. But it doesn't need to know that just so we don't confuse people. So moving up those log evidence gradients very much like,

you know, paradoxically moving up concentration gradients to sort of clump together to keep in this attracting set and to resist the dispersion of the random fluctuations inherent in these random dynamical systems is just the same as moving down the negative log evidence, which is the energy itself. It's the same thing. So what does that mean? Well, what it means is the generative model is something that you bring to the table to endow the behaviour of this thing with an explainable teleology of the kind. It looks as if this thing is acting in a way to maximise the evidence or gather the evidence for its generative model. And this is what I think its generative model is.

But notice the thing in and of itself doesn't go through this process.

It could, of course. So if I now, people, things like you and me, we do have a model. We do have and we communicate it and we talk about it. But most things, you know, most things would not possess that that deep structure. So you can invoke homunculi, you can invoke self-modelling, you can invoke you know, the kind of, not homunculus in the sense that you would normally end up with an, you know, with an infinite regress, but you know, the kind of metacognitive aspect that comes from any deep or deeply structured hierarchical generative model. You know, you can get that sort of homunculus like detachment and metacognitive perspective on things. But strictly speaking, you know, the generative model is only entailed by the dynamics. And I should say that, you know, this is probably best to think about, you know, applications of the free energy principle in, you know, in, in, as being limited to simulating and modelling and understanding other things, as opposed to oneself, unless one's doing self-modelling.

And so, so in that sense, you are ascribing a teleology, a purpose, a purpose that would explain the behaviour of something that you're observing. But notice, because you are observing it from its point of view, you are an external state. And from your point of view, all you can see is the Markov blanket, you cannot see what's going on on the inside. So it is, it will be mathematically impossible for you to ever know whether what's going on on the inside is indeed described as even a gradient flow on a free energy functional, let alone, did it have its own internal model, did it have its own beliefs, it just looks like that from your point of view, because you will never, ever know, by definition, if you knew, then you would, by definition, have breached the Markov blanket, and that thing would cease to exist as the kind of thing that it was before. I mean, it may sound obvious, or it may sound trivial. But I don't think it is. If you just look at people like me, who spend our entire lives trying to peer through the Markov blanket using brain imaging, for example, and, you know, or people doing comparative anatomy by dissecting dead things, or by inventing new techniques to try to try and get, you can never get beneath the Markov blanket, but you can certainly sort of have certain perspective on the Markov blanket that are closer and closer to the internal space, but you can never actually get in there. I mean, you know, there's a, I imagine a whole philosophy on this, that you, you, you couldn't see your own visual processing, you couldn't hear your own cochlear dynamics, you know, you, you, you can't ever have access to the inside, very much in the spirit that internalists would say you can never have direct access to the outside. So I think that also applies to these stories about generative models and the Bayesian mechanistic interpretation of, that comes along with the, with the free ownership principle. Does that help in some

way? It's a little bit subtle. And I can just imagine a lot of my friends tearing their hair out because they have very particular views about this. You know, people think it's, you know, it's a model of a model. Some people think it's, you know, take a very internalist, some people externalist. Yeah. Mathematically, you know, it's, it's not the model which drives, drives us, you know, it is the, it is the dynamics which drives us. And that dynamics can be interpreted in relation to a generative model, should you want to. Excellent. Well, I'm going to follow that philosophical line of thinking. And what I want to ask is, this is an interesting point, that the notion that we can get to a closer approximation of what the internal states of ourselves are, for example, but it's always going to be an approximation because piercing the Markov blanket would constitute self-disintegration under the, under the rules of, or the principle, the free energy principle per se. How, how in a sense, therefore, do we know, or how, how, how, how much can we even make the claim therefore that there is such a thing as an external dynamic that goes beyond anything, goes beyond our sensory inputs? It may seem like a deeply non-scientific question, but you, the way you were speaking couldn't, I couldn't help but think of people like George Berkeley and even Kant and this phenomenal noumenal distinction. If we don't ever have direct access to so-called external dynamics, because that is part and parcel of the, the physical game that we're in, why can't we just say that there are no hidden states and all we get are probability distributions over sensory appearances, but we don't need something like a likelihood mapping to something that's latent? Well, you, you, you could say that and a lot of my friends do say that. And I think in the sense that the notion of direct access means that you will never know in some sort of heuristic sense what the external states are if they exist at all. And, and, and, um, um, on the other hand, you could also argue, um, that, um, there is profound access to the, um, um, uh, uh, and existentially meaningful access to the external states. It's just organized in such a way that it, uh, it has to be transacted by the Markov plan blanket. Um, so, you know, I haven't, I haven't had this conversation for, for, for, for several years now. So I used to have lots of pat answers, which, uh, uh, depending on who I was talking to, but the, you know, the way that you phrase the question, which was very clever, um, just made me think, well, you know, what is the internalist argument? Does it mean that because I don't have direct access, that the external states don't directly influence me, that means that they don't exist? Um, is it, you know, is that the, the argument that an internalist would make? Well, may, might it be the case that all we have, we don't need to invoke a kind of further realm, a further causal realm, an underlying ontological realm, if all we have are just the streams of sensory input and seemingly some regularities, some patterned regularities in them. Why, at least in terms of doing the Bayesian modeling, do we need to invoke a hidden state, uh, an underlying cause? For example, like, if we just take that, if we look at the low road to active inference, we talk about how human, if we talk about how cognitive creatures like ourselves

predict the world, let's say at a very simple level. I, a question that I've always had is why do we, why does the claim to predictive processing or predictive coding more generally, why does it make claims about hidden states? Why doesn't it make claims just about regularities within streams of appearances? That's the way that I would pose that question.

Right. Um,

so I'm just mindful you've introduced, uh, predictive coding and predictive processing. We have the energy, uh, perhaps it would just to, just to, uh, for those people, um, who may not, uh, know that the, you know, the intimate relationship between all these things. So another way of looking at free energy is, is the, uh, amount of prediction error. And more specifically, if you're talking to some people, it would be the, uh, the total amount of precision weighted prediction error, um, which means that there's another reading of minimizing free energy, which is not the maximizing the model evidence, but, minimizing surprise where surprise is the implausibility of some sensory data, um, given your model of that data. So in exactly the same way that if I go around gathering data that I find, uh, continuously surprising, um, then that's basically evidence I haven't got the right kind of model to be, um, to be able to explain, i.e. predict those, those data. Um, so again, we have one of these, um, um, sort of completely equivalent descriptions, but, you know, they have very different, uh, flavors that to maximize model evidence is to minimize surprise, which is to minimize a prediction error, which is to maximize predictability, which is to minimize variational free energy and so on and so forth. They're all the same. They're all the same thing. So when you're talking about predictive coding as a particular, um, algorithm for minimizing free energy, uh, technically, um, you know, a variational scheme, which is known in engineered as a common filter. Um, what you're saying is I'm making certain assumptions, um, about my generative

model, um, that means I can write down these free energy gradients as prediction errors. So literally the prediction error, my apologies, the precision weighty prediction error literally is mathematically exactly the gradient that we were talking about before. Um, so just, uh, this is, you know, join the dots. And so, anyway, to your question, um, uh, why is it that, um, I need to invoke latent states in order to make sense of data? Um, I would argue, um, that the sense making is just explaining data in terms of latent states. Um, um, um, those latent states are often referred to, um, you know, in the technical literature as, as hidden states. And what do you mean by that? Well, it just means they're unobservable there from the point of view of the free energy principle, they are hidden behind the Markov blanket. So I would say that detecting patterns simply is invoking latent states that organize and structure those patterns. Um, so, um, I don't think there's anything magical about, about latent states. So the latent states that are, if you like, entertained by your internal dynamics and your sense making, um, uh, are not the external states that, you know, that they are descriptions that are only relevant to the generative model that, um, you need in order to define the prediction errors or the free energy gradients in order to, uh, you know, simulate or describe, um, your sense making in terms of your neuronal dynamics or your internal dynamics or the operations of say a thermostat. But there's a twist here. Um, um, um, there's a reason it's called active inference because it's got a big active at the front.

Um, so it's not good enough just to say, oh, I can understand psychology just as making lots of sense of my sensory impressions. That's, that's not good enough. Um, because you know, you are, um, um, coupled to the world, um, in, in, uh, in, uh, in, uh, in, uh, in reciprocally, which means that you actually, um, are not only are you exposed to the world, but the world is exposed to you in an exactly symmetrical way. And what would that, um, you know, how could one describe that? And then we come back to, uh, where we, where we started with this partition of blanket states into sensory and active states into the inputs and the outputs, um, which means that sense making is quintessentially active. So one way of phrasing that is that, yes, I'm making sense of all this sensory data and all the patterns. Um, but at the same time, I'm actually in charge of actively soliciting those data, causing those patterns. Um, so I think once you, um, once you bring that sort of inactive, um, aspect to the table, so notice, um, I noticed that you, you, you, uh, treated predictive coding as a, as a sort of, um, generalization, predictive processing. You might be easier if you did the other way around, because then you have the grace to, um, accommodate action in predictive processing, it's most general sense. And if you don't, then predictive processing, you know, I think is an incomplete story and you have to have, um, as part of the processing, the active solicitation, the, um, the, the garnering and the structuring and querying the world in a way that reciprocates with the right kind of information that allows you, to do the good sense making in terms of, um, you know, the latent states that we're, that we're talking about. Um, I mean, formally, and this takes us, um, into another part of the story, which we, we, we, we didn't previously unpack about active inference. So somebody might ask, well, okay, so we've got this, um, um, application of a, um, a free energy principle that a physicist might be quite comfortable with, um, um, to, um, behavior that entails some kind of sense making, um, cheekily with, you know, I, I like to call this sentient behavior, and then I get told off for that. Um, but, um, so, uh, so how does this differ from, um, reinforcement learning? You know, how, how, how, you know, behavioral psychology? Um, it differs in a quite fundamental way, um, because the, the big thing that active inference brings to the table is that, um, first of all, it says you are in charge, you are actively, uh, sensing, you know, it says this is active vision, active perception, active inference in, in its most general sense, um, which means now what are the imperatives for generating your own data? Now you could, if you were doing behavioral psychology, uh, you know, uh, of, of a behaviorist sort in the 20th century, you could say, well, I just want to generate those sensations that I find rewarding. Um, and so I've got some, privileged sensory, um, channels that, um, I'm going to label as, as reward, and I'm just going to act in a way that maximizes the reward that comes in. From the point of view of the free energy principle, that's not what, that's not what happens. What happens is I want to find those data that resolve the expected surprise or the expected free energy consequent on acting, soliciting those data. So what's expected surprise? Well, technically it's an entropy, it's just a measure, a mathematical measure of uncertainty. So what that means is that the, um, the, the description of things that persist or exist over some particular period of time, um, reduces now to a kind of Bayesian mechanics in which the behavior will look as if it is information seeking and uncertainty resolving. Um, it will look as if it is responding to these epistemic affordances under constraints,

and it is those constraints that, um, um, if you like, would be the homologue of the reward.

Um, they are literally the constraints in doubt by the prize on the generative model, the kind of, um, sensory states that, that constitute this attracting set, you know, to which I am attracted. But the, the vast majority, um, of the drives for, for good, um, good behavior, or sort of behavior that things that exist would evince, um, is this resolution of uncertainty, this information seeking. And indeed, you know, um, that leads you into considerations. Well, what, what is the, how would you describe that kind of information, that optimal, uh, information seeking behavior, um, if you were a scientist or, um, if you were an engineer, and if you were a scientist, you'd be looking at the principle of optimal Bayesian design, um, articulated by people like, uh, Dennis Lindley, uh, in the middle of the last century, um, where you measure the quality of an action in terms of the information gain afforded by the data that, uh, you secure by acting in this way. Exactly the same idea, um, um, emerged in the context of active learning in machine learning. Uh, you know, this is the problem. If getting data is costly, what data point would I need to resolve the greatest uncertainty about my, um, beliefs about the latent causes of those data? Um, so active learning, again, notice actives, uh, your actives, um, your, um, an explicit part of this process. So in answer to your question about, you know, would it be sufficient just to understand sentience in its most basic or elemental, um, form as discovering patterns in data I would say no, um, because that denies the openness of any given system immersed in her world, um, in two directions, you know, the world influencing you and you influencing the world. And as soon as you put that into the mix, I think it's very difficult to conceive of any construct, um, possibility in philosophy, um, but certainly any construct in computer science or, um, um, uh, or physics that would, um, um, that would permit a complete description of sentient behavior just in terms of sensing patterns.

Does that make sense? Yeah. Yeah. That makes sense. That makes sense. Um, makes sense to me. Um, yes, I want to stay on action for a second because I think the, the notion of, uh, the minimization of free energy or the maximization of model evidence makes some, uh, could make some intuitive sense for an agent like ourselves in the sense that we, we have our sensory inputs, which are divergent from our priors and we act to change the world to make those, those sensory outcomes more in line with our expectations, our preferences. However, at a more fundamental level, uh, one, one definition that we spoke about, the free energy principle is that the internal and external states look like they're tracking one another across a Markov blanket. Now, a question that comes to my mind when there's, when I see a definition like that, which is, I, that makes sense for me, uh, for a sentient, conscious human, but what would a stone be tracking or what would a drop of oil be tracking to use an example that you've, um, used before? I think so using, going to something a bit more less animates, and I presume the answer lies in animacy. Um, I think would be a really interesting avenue to go down because it gets these fundamental questions of how does something as minimal as a particle retain its particle-ness and, and what, and what does that really look like when, when we don't, when we presume that they're not conscious unless we're panpsychists, but we won't go just down that

route just yet.

So, well, that's where you were taking us, which, which is a fun place to be. Um, right. How does it, how does it, um, elude panpsychism in the free energy principle? Well, I think you've, you've just, you've just, uh, uh, answered your own question there. So perhaps we will just unpack that a little bit. Um, yeah. So what's the difference between a stone, um, and a thermostat and, and, and me? Um, and I think what you're, what you're now asking is, um, are there, um, what I will call natural kinds, but I wasn't allowed to because that has a lot of philosophical baggage, but for the purpose of this conversation, I'll just say, are there natural kinds of things, um, that display different kinds of behavior that, uh, coming, that can be equipped with different kinds of teleology? And I would say absolutely. Uh, and you've just rehearsed, you know, the, you know, the, the, the, the, the, the key things. So yeah, as you noted, um, to be animate, um, is a big thing, um, um, in the sense that the stone is not animate, um, uh, in the sense that it moves itself. You know, if stones started, you know, wandering uphill or flying around, that would be much more interesting. Um, but they don't have that animacy that is characteristic of biotic self-organization.

So what does that mean in a sort of, um, uh, deflationary sense of the free energy place? It just means that the active states are an empty set. Um, and usually, um, for stones, you might also argue that the internal states are empty. We don't have to, you know, the stone can be, um, you know, making sense of its world in terms of its latent states or can acquire that into interpretations. Perhaps that's a nice example of what we were trying to, um, get, um, drill down on earlier on in terms of, um, is there a generative model? I mean, I, one could argue that a stone has a generative model of external milieu in terms of the temperature and the latent variable, which is a hidden state, it's a latent state, you know, as well known in statistical thermodynamics, which is a sufficient statistic of the distribution of things. And the temperature, um, could well be, uh, inferred by the internal, the interior of a stone. And therefore it is making sense of the pattern of its sensory exchanges with the world on, on the surface of the stone, simply because it's warming and cooling and therefore tracking the ambient environmental temperature. So that's perfectly free energy principle consistent. Um, it's a very boring kind of, uh, of sense making, but you know, and it is that boring kind of sense making that I had in mind when talking about, um, if you preclude the active part of active inference and just think about sort of perceptual inference, um, then, then you're sort of missing the point really. Um, but I think a stone is a good example of that. Now, would you say the stone has a generative model? No, you could, you could argue that it, but, uh, you know, it's you trying to, trying to, um, if I explain the behaviour of a stone, but the stone doesn't have very much behaviour. Um, but you could, you could start to think, well, now I can test a hypothesis. It is actually registering, um, uh, the, the temperature on the inside. And again, just to rehearse the argument, the fundamental argument we were talking about before, you will never know though until you break the stone, because once you break the stone, it's no longer a stone. It's just a broken stone. Uh, so you're exactly the same problems or issues confront you even with the stone, but you're focusing on, on that sort of big move from things with and without active states. I think then you're talking

about things that are animate, things that have, um, animacy and act upon the world. Um, is that sufficient to get the kind of behaviour that we've been talking about, which is, um, you know, the kernel of active inference in terms of, um, what active inference brings to the table in terms of, information seeking and having a variety of different affordances, particularly epistemic affordances, um, as an explanation for, for, for, um, for behaviour. Um, you could argue, no, you know, a thermostat can act upon the world. It can switch heating elements on or off and it senses things by its, by its thermoreceptors. Uh, you could say like, very much like a Watts governor, um, it, you know, it now has active states. It has inputs and outputs and it has internal dynamics and, um, um, and in one sense,

the Watts governor is a good regulator and therefore conforms to, um, the cybernetic view of, of, of, of, a generative model and it's acting in the right kind of way. So does this kind of system, um, have that the curiosity that would be associated with information seeking and resolving uncertainty? And you would argue, well, no, it doesn't. So now you've got another kind of natural kind, um, which does. So what's the difference between, uh, you know, a Watts governor or a thermostat or possibly a virus, um, and me? Well, I think the key difference is a particular kind of, um, self modeling that just comes from being very big and a bit structurally complicated in the sense that there are certain parts inside my body and inside my brain that are so distant from the active states that are actually moving, um, my arms, my actuators, or, um, are engaging my autonomic reflexes that I cannot see them directly. Um, and if that is the case, the only way that I now can sense the consequences of my action is vicariously through the sensory states. Now this introduces a really interesting distinction between me and a virus. It means that I now will start to treat my own action as a hidden or latent cause of my sensations. And that brings to the table, okay, well, if now my own action is being inferred as a random variable, then now is a separation between what I think I'm doing or what my, uh, active inference, um, um, would, uh, allow me to talk about in terms of what I'm, what I'm planning to do or what I'm inferring to do in the spirit of planning as inference and what I'm actually doing. So this, um, brings you to an even more, uh, a different kind of thing, which would be something like me, um, that now has beliefs about, um, its own action, um, which is distinct from the actual real variables, you know, um, that constitute the active states of the Markov blanket. Um, and as soon as I have beliefs about my own action, I have to ask myself, well, as a physicist, I would ask, well, what, how, what, what, what are the probability distributions over my, um, my active states, my, um, those states that, um, actually change the causes or the, um, the, the external states generating my sensations. And when you write that down, um, what you get is exactly what we're talking about before, which is this mixture of expected information gain, you know, basically expected free energy that has, can be decomposed into this information seeking part and the constraints afforded by your prior, uh, the prior part of your generative model, the things that you find very surprising. Plus it is to unpack that, um, um, intuitively, you know, let's take the predictive processing surprise minimization view of, um, self organization and, um, an active inference. And that means that I now have the, I am the kind of creature that exists. Therefore I must be minimizing my surprise. Furthermore, I now have beliefs about my

actions. I have prior beliefs. I must have prior beliefs about my actions. Of course, this is all sub-personal. So what kind of priors would I have about my, my own actions? Well, I'm going to, um, because I exist and I'm a, um, a free energy surprise minimizing kind of thing because I exist, then I must choose those actions that minimize the surprise I expect following that action. So that means I'm going to minimize my expected surprise. Now, there are two ways in which I can do that. And we've already spoken explicitly about the two ways. One way is just to notice that the average surprise, the expected surprise, um, is just uncertainty. So I can reduce my expected surprise or my anticipated or my, um, average, surprise, um, by, um, getting to grips and knowing, you know, what would happen if I did that, uh, with greater precision. So this is information seeking. This is the sort of the curiosity. Um, it's the novelty seeking part. It is the, exactly the part that underwrites the, um, the, um, base, the principles of optimum Bayesian design, getting the data that minimize my uncertainty because it certainly just is expected surprise. And there's another way of doing minimizing expected surprise. If I know what is very surprising, if, you know, by being, um, um, at very, very low temperatures or being very poor or being snubbed socially, um, everything that is, if you like, not characteristic of me, given that I am, you know, and I exist in a, you know, in a particular way, um, then any deviations of that basically, uh, can be thought of as surprising. So if I stray beyond very much in the spirit of, um, homeostasis, although now we're talking about allostasis because we're talking about the future, um, then I'm going to choose those actions that don't just, uh, do the information seeking responding to the epistemic affordances, but also will elude those kinds of surprising states that are very uncharacteristic of me, being very poor, missing out on that opportunity, uh, being embarrassed, uh, being in pain. Um, although paid is probably a better example. We're talking about the sensations that we want to, uh, want to avoid. So, um, that would speak to this, um, um, the prior, um, the prior cost or the prior surprise, the negative, which was, was the preferences you met, you mentioned before. So, um, you can look at these constraints afforded by the priors, the pragmatic part, if you like, of, um, of these affordances. Some people call them instrumental, so you've got epistemic and instrumental, um, affordances that just are the expected information gain and the negative expected cost, which would be the, um, you know, uh, manifestation of the constraints. You can look at the, the complement of constraints in terms of where I don't go in my sensory state space, um, as, um, where I do go, which are my preferences, my preferred characteristic attract attracting sets. So, uh, just to remind myself, and you, and people who are listening still at this stage, um, why are we going through all this? Well, because this is what comes out of a consideration of beliefs, prior beliefs about the way active state should unfold, uh, how, uh, and that becomes very pertinent when now I have to actually, um, uh, embody those beliefs in my generative model. So suddenly now I become something that is very distinct from a thermostat or, or a virus, because now I have a generative model of the consequences of my action, and that generative model is very simple. It just says a priori. I will, um, a priori think that the action that maximizes information gain and maximizes this instrumental value, uh, are going to be the most likely policies that, and this, you know, could, uh, could be construed as a, um, a high-end kind

of planning as inference. It's not as trivial or simple as a sort of, uh, KL control because we've got this epistemic, uh, part, part of it, but it still has a spirit of planning of inference. So what I'm saying quite simply is there are certain things that plan and there are certain kinds of things that don't plan. A virus doesn't plan, the weather doesn't plan, evolution doesn't plan, the stone doesn't plan, and you could argue that many, um, um, smaller insects don't plan, but as you get bigger things, then they start to plan. And I think you move from non-planning to planning, at which point, by definition, you are ascribing to these bigger things like you and me, um, a, um, ascribing, um, a teleology to our behavior that rests upon a generative model that includes the consequences of its own action, which crucially equips it with a temporal depth because the consequences are in the future. So unlike the thermostat that doesn't need to look very far into the future, things that plan that are so big that they, um, lose contact with their actual actuators, um, now look as if that they have this sort of temporal depth, um, this, this temporal thickness to their generative models, specifically enabling them to, uh, plan, uh, plan their actions in, into the future. So that's how I would get out of the panpsychism argument.

Okay. I've never, I've never heard that argument before, um, about the distance from the actuators. That's very interesting. Um, because it seems in that sense that the temporal depth is downstream on size in a sense. Um, is that, is that a correct reading and, and is there a reading of your argument that you've just laid out there in which you can have deep temporal model, uh, modeling planning as inference and, um, allostasis as well as a sense of, um, retrospective inference without being a large thing. Is that, or is this notion of being, uh, far from your actuators so that you have to start disambiguating what is your action and what is the action of the world or other people?

Is that fundamental? Um, yeah, I mean, this, this is a really interesting notion because I've, I've actually never heard it really be, uh, elucidated like that. So I'd love to know whether there is a way that a virus could be doing deep temporal modeling despite its minimal size.

Right. Yeah. I mean, uh, to be honest, I don't know. Um, but a, a, a sort of, um, an intuitive answer, uh, would be, uh, no size really does matter. It really does matter. So by size, I, I, you know, I'm implicitly talking about the, um, the, the, the sparse coupling that underwrites the conditional dependencies in modern dynamical systems that give rise to Markov packets and blankets and blankets. So, um, I'm not literally talking about, you know, how, how, how wide something is in terms of millimeters. I'm talking about the sort of, um, the hierarchical depth of, um, conditional dependencies and being secluded behind Markov blanket after Markov blanket after Markov blanket. So, um, if you, if you're comfortable associating, um, physical size with that, um, hierarchical depth in terms of, um, what a thing is, you know, in terms of having this internal hierarchical structure, internal Markov blankets, then I think the answer is very clear. No, you have to have, you have to be sufficiently big. So I'd be extremely surprised. Um, so for example, I haven't said this before, but, you know, I would imagine very, very small insects can't plan, whereas, um, something the size of a bee might be able to plan. So I'd imagine that a *Drosophila* couldn't plan, but I, I would imagine a bee might just be able to get there. And it, and it, it starts, um, you know, first of all, that, that, that immediately speaks to

a philosophical vagueness in terms of that temporal thickness, which, which is interesting. I mean, is it, is it categorical? You know, can I say this kind of artifact, um, can be explained in terms of a generative model that does not cover its own action because it has direct access to its actuators, because it's sufficiently simple and small. There are no Markov blankets that intervene between the, um, the, you know, the, um, the, um, effectively the, the, the sense-making part, and the, um, uh, um, for example, sending predictions and the predict-impressing spirit down to the, um, the actuators. Um, whereas this thing is so big that there are inevitably with probability one, so many, uh, Markov blankets intervene, there is no now direct access to action upon the world. Um, and, um, so that might be a, uh, a bright line. It might be a, um, a qualitative distinction between things that do and do not have, um, uh, a representation of their own actions. Um, on the other hand, the counter-argument, I think, would be this more vague notion that, you know, even, even a thermostat could be construed as having a little glimpse of the future simply because it could be, um, uh, cast as, say, a PID controller, which basically, um, um, invokes now rates of change. And as soon as you invoke rates of change, you've got sort of, you know, a mathematical image of a little trajectory into the future.

So it could be a great, it could be a graded thing. I've lost the, I've lost the importance of your question. We'll ask you a question again. It was probably, it was probably unimportant. Um, but this is really, sorry. I'll go again. This is a very interesting line. I'd love to, I'd love to dive down into. So what, so my interpretation of this, my, my reading of this is that in a sense, having bundles of Markov blankets embedded in Markov blankets leads to the inevitable inference that there is something like me, which is doing something. And there is something that's not like me, which is also causing sensory, um, data that I'm receiving. That makes sense. Now let's, let's tie in temporality to this. Does it, does this, does this mean that in a sense, temporality is a corollary of selfhood in the sense that the selfhood is almost more fundamental than the, the temporality because it comes from the dimensionality of the things that are Markov blankets upon Markov blankets and how can we integrate temporality and temporal thickness or depth, um, into this picture?

Right. That's absolutely right. And, um, you could tell the story in a number of different ways, you know, if you're talking to, um, for example, Chris Fields or Maxwell Ramstad, um, they might start talking about, um, irreducible Markov blankets, um, that necessarily, uh, require the notion of nested Markov blankets and that there is some one or more core irreducible Markov blankets that would have, um, um, this aspect of looking at once, one's own inference processes and acting internally. And you get the notion of mental action. You can then start to work towards, um, sort of qualitative experiences in terms of self modeling and what that might look like. Um, the other, the other way that you could tell this story is to know that, um, to invoke, um, the renormalization group, if you were a physicist and just know that when you're talking about any hierarchical generative model, there is, um, inevitably a core, a core screening that includes time in that, in that depth, um, which means very simply that the deeper parts of my generative model and indeed viruses are on a free energy principle view, um, um, encoding beliefs

about things that change more and more slowly as you go deeper and deeper. So there's a separation

temporal scales. Um, so yeah, I, I, I, a commonsensical view of this is that, um, the, the neuronal representations, uh, down near my primary auditory cortex or indeed in, in, in, in the brainstem, um, are representing very, very fast fluctuations in, um, pressure and then frequencies internally, uh, and, and, sort of, centenographic-like, um, um, representations that are changing, um, you know, possibly even in fractions of a millisecond if you're an owl, um, uh, but certainly over, over 100 milliseconds.

Then I move up to sort of, you know, uh, primary auditory cortex and I may be in the realm of some, um, um, the, um, hundreds of milliseconds and sort of frequency glides at defined phonemes and I move up to secondary auditory cortices and then through the auditory hierarchy, getting to the level of, um, um, words right up to, um, say, you know, some parts of the prefrontal cortex where we've got the entire semantics and syntax and narratives and stories start to emerge. And as every time we go deeper into the hierarchy, we slow down or we extend the temporal compass, thereby providing a context for the faster influences and fluctuations at the level below. So again, we come to this, um, you know, notion of, um, size entailing, um, a, um, a certain kind of depth in this instance, a hierarchical depth in the generative model that, as you say, necessarily goes in hand in hand with a temporal depth and a separation of temporal scales. So I think you're absolutely right. There has to be a temporal aspect two deep generative models and by deep, I just mean that there are nested Markov blankets. So, you know, to define a hierarchy is only defined in terms of the conditional independences, um, and implicitly the Markov blankets of one level in the hierarchy, um, um, that defines it as a level in the hierarchy. Without the Markov blanket, the hierarchy would not be there. So that's another example of this sort of, um, um, being open but being closed, um, um, um, being able to individuate something from something else. And this is the thing is just to the level of a hierarchical, a hierarchical generative model that we're ascribing to, um, the dynamics of some internal, uh, of some internal states. I think there's a third story you could tell, which is possibly where you're going, which is, um, um, how it relates to, to consciousness, um, um, in the sense, and this is a very simple story that, that, that, um, if I am, um, forming beliefs and doing, um, belief, basing belief updating as prescribed by active inference about the consequences of my actions, then that has to be in the future, as we've just said, and perhaps certain things think deeper into the future than other things. Um, but notice what we've done here. It's about my actions. So not only have you committed to temporality, um, and temporal depth and temporal thickness as an attribute of things that plan, um, um, and, uh, more like you and me, but you've also committed to agency. So now you've created from something that was previously not an agent into an agent where I'm using agency in the sense that I, one would not say that a thermostat had agency, but you would say that I have agency, um, in the sense that I have beliefs about the consequences of my action and I, um, will, or can be described as, um, using those beliefs to select a particular action. And, uh, you know, coming back to predictive coding formulations that, that selection would basically, I'm going to predict the state of mind, um, multi-planet or my autonomic, um, nervous system, um, provide it with the right set points. And, and then the, the active states will actually reflexively fulfill those prophecies, those

predictions, um, and everything is working properly. I'll, I'll be a good allostat or in Roshmiz terms, a good homestat. Um, so what, what you've done there is, is, is bring agency into the game. Um, and I think that, again, this speaks to the inactive part that, you know, you now, because you've got beliefs about the future and specifically the future that you have caused, that you have authored, you have now become an agent. So we have, I think, got this, at this stage to consciousness and self-awareness, but I think we've got the necessary foundation that there are only certain things that have to be agents, I think, to be conscious of themselves as agents, as doing things, um, as things that exist in, um, um, in a world, uh, in the sense that their existence is manifest in terms of the consequences of the way that they act upon that world. If it wasn't, you wouldn't need any of this, you wouldn't need this temporal thickness, you'd just be, you, you know, you just need the apparatus of a stone or a thermostat to make, to do, to do your sense making. But as soon as you've got agency in the game, I think then you've got, you've got something which takes a step closer to, um, having a very deep and possibly it's this irreducible Markov blanket, um, um, um, there of the generative model or level of a, of a deep generative model, which actually now starts to entertain the hypothesis. I am an agent, um, which you don't need, but you could, and you have to ask yourself, why would you ever want to entertain that hypothesis? I am an agent. Um, so this would be minimal selfhood, just a representation, um, that I can be in different states of mind when it, uh, that, uh, and those states of mind are something that underwrite or I can condition on in terms of what I'm going to do. And that doing could be internal. It could be how I deploy attention in terms of precision weighting in predictive coding, um, so it could be covert action or it could be overt action. I could now, um, select, um, you know, um, certain plans at a lower level that would be then, um, enacted by my motor system or my autonomic system. If I was that, that kind of creature, does that make sense?

It makes sense. And it's opened up a can of worms in my head. Um, well, inevitably, because we bought in lots of concepts here and I want to try and unpick them, um, for people at home and for myself, but it's fascinating. So I'm loving it. So let's keep going along this, along this road. We've got agency, consciousness, and temporality as, as these three things I want to have a little look at. Starting with the first thing we talked, uh, you, you mentioned as a response to my question about temporality, that we have a deep hierarchy in which, um, more deeper levels of the hierarchy are tracking, um, slower fluctuations in the external dynamics and therefore contextualize and constrain the higher order, faster dynamics. Is it, are we in this model, are we invoking time as a fundamental aspect of those external dynamics or is it an inference that the act, that the, um, that the organism is making about the nature of reality that it's experiencing? Um, well, that's, that's, again, an excellent question. It certainly, um, does not, um, um, it does not address time perception, uh, you know, in terms of the psychophysics of time perception. Um, does it address, um, time in a more fundamental sense? Um, I, I don't think it does, um, uh, in a sort of, um, sort of folk psychological sense. Um, it's really, um, um, so the, the, the, the answer to that question normally, um, is framed in terms of relative movements in a belief space. Um, and, um, so if you think of time as, um, part of some metric space

space time, then what you're talking about is time as something that can be measured, time that has a, a distance, a milliseconds or seconds of, and the like, um, could me, um, could, uh, me as a generative model,

um, um, ever have access to or build, um, hypotheses, uh, of time as a metric? Yes, we certainly could. Um, however, um, it would be very difficult, um, you know, ignoring sort of internal clocks and the like. Well, and actually, no, you don't have to ignore internal clocks. You then ask, how do you do it?

And normally what, what the story, the story is, um, it's the, the number of moves you make at one level relative to another level of a deep generative model. And the number of moves is quite crucial here in terms of tying that down mathematically. Um, and my favorite way of thinking about the number of moves is basically something called the information length. And the information length, um, um, refers back to something we've mentioned before, which is information geometry. So as if you think of our, um, so one picture of the free energy principle, uh, it's quite a technical picture, but I think it's quite useful, um, is that, um, our internal, say brain states or the internal states of a thermostat, um, encode a probability distribution, a Bayesian belief, a conditional belief about some latent states, um, which means that for every state of the brain, there is, if you like, a point in the state space of all the physical states of the brain. But each point in that space now stands in for a probability distribution. And that means that, um, this is a special kind of space because it is equipped with a metric. What is that metric?

Well, it's the KL divergence. Well, the path integral, uh, infinitesimally small KL divergences as I move through this space.

So this is what, this is what defines information geometry. The information geometry is a very particular, um, bit of mathematics, um, or theorizing that applies to very, very special state spaces or manifolds, where every point in that, on that manifold corresponds to a belief or a probability distribution.

So now, what, if the metric is the, um, the KL divergence or, you know, the information length, which is, uh, uh, related to the, um, the accumulated KL divergences as I move from one point to another.

Um, what we are saying is that the, um, the number of moves I make, the number of units as measured with the, the, um, this kind of metric, this information geometric, uh, measure.

Um, which actually technically is this Fisher information, um, is, is, um, basically, um, a description of, um, the precision by which I measure whether I've changed my mind.

So if we say that I've made n moves, that means I've changed my mind n times.

And I know that because I can measure the information length and I've actually moved a significant, uh, by a significant, um, or by a measure that is large,

simply because it is larger relation to the precision that, uh, um, that, um, that, uh, that provides the distance between any two points on this.

So what that means, to put it very simply, is that time can be measured by the number of times I change my mind at level n in my hierarchy,

divided, divided by the number of times I change my mind at the lower level.

So now you've got a relational, uh, you can gauge time by the relative degrees of belief updating, uh, you know, um, the information rate, which is the, you know, um, um, um, um, um, um, confusingly, the, the, the, the rate of change of the, of the length with, with time.

Um, but where time is now, uh, contextualized by the level above or, or by the level below.

So you have this very relativistic view of time, um, and you could have different levels, you know, I would imagine, um, if you, uh, you attend to different levels of your hierarchy, you could get a very different kind of time perception.

Indeed, we know that attentional set and, um, um, um, in conditions where, um, say taking psychedelics or, uh, in things like Parkinson's disease,

where, where you get, um, changes in the neurobiological encoding of uncertainty and therefore this Fisher information metric,

you can get distortions and different senses of time.

Um, um, um, um, so what I'm saying is there are probably as many different senses of time as there are levels in a hierarchical model.

And they can only really be measured in, in, in a relative sense in terms of how many moves do I make at one level in, for any given move at another level.

So, um, that, that what we've done now is if you like, um, take temporality off the table to the extent now that we're down just talking about, uh, movements in an information space or a belief space that has this kind of information geometry.

These are really fascinating issues.

I think, uh, I think yet to be fully explored, um, my, my friends, uh, um, um, has done probably some of the best work in, in, um, in this area.

Um, but you know, it, it is, it is to my mind, um, a really interesting issue because, you know, it really forces you to think about the, um, what you mean by separation of temporal scales and what, you know, what, what kind of course draining over time is implicit

in this kind of move from one level to, uh, to the next level.

Does that address your, what you were, what you were?

Yeah.

I mean, as I was saying, this is all really interesting.

Um, and again, something I hadn't thought of, um, before in terms of, in terms of this, in terms of this contextualization process being about moves, which implies to me that, but there has to be, it's like, this sense of time is contingent on these moves that are being done at the different levels of the hierarchy.

Does that process itself not imply some dynamic, some temporal dynamics?

How can one distinguish a move of 10 relative to a move of five without having segmented moves in some kind of space time geometry?

Yes.

Yes.

And it is exactly that, um, segmentation that appeals to the information geometry.

Um, so yeah, how would you actually do that segmentation?

How, how would you measure, um, the amount of belief updating, um, in universal or clock time, um, at one level of a hierarchy?

And, um, this is where the information geometry comes in.

It's basically the number of different brain states that you have entertained during your movement on this

statistical manifold, which is also, you know, I think very nicely articulated in terms of belief updating.

So you can literally think of this sort of, um, the space that would be traced out if I were to plot, um, the activity of all my neurons, um, on every axis of some very high dimensional state space.

That would be a statistical manifold.

My belief updating, which is a continuous process.

My gradient flow, um, is literally moving on this manifold.

The distance I move is the information length that literally scores the degree to which I change my mind.

It's also mathematically the information gain, um, as measured, well, closely related to the information gain as measured by, um, the KL divergence between my prior and my posterior.

After I do my belief updating and I'm moving on this manifold.

And the question is, you know, how far have I moved?

Uh, and can I segment, segment that into units?

And that's basically what the information length, um, uh, you know, allows you to do.

So I, I, I love this picture because, um, if you just think about what kind of flows on this manifold would, um, be prescribed by the free energy principle.

Because the free energy is a functional of the, uh, beliefs, um, encoded at each point.

Then the free energy is a landscape on the statistical manifold.

So we are perpetually falling downhill on this manifold.

The free energy gradients are themselves and the free energy is itself changing because my beliefs are changing.

So I've got this lovely itinerant, you know, model or picture in mind where I've got a statistical manifold.

It is now, um, where I have gradient flows where I'm flowing on this manifold, um, using a gradient flow on free energy where the free energy now provides a Waddington like landscape chasing minima all the time.

And of course that chasing reflects the fact that the free energy itself changes as you move around the manifold because your beliefs are changing and the free energy is, uh, a functional belief.

Um, and then you introduce this, uh, this notion, which I think is really important.

Um, the notion that the high level of a deep model, um, provides the context.

And what would the context look like in terms of this free energy landscape?

Well, it means that the landscape would change.

So, you know, I've got this gradient flow, which is, um, um, from, um, on a fast timescale, just flowing down the landscape, down the landscape.

But at the slow level, the landscape is itself now changing all the time.

So I'm perpetually chasing minima, um, and moving minima because the gradients themselves are contextualized and are changing slowly.

So you get this deep structure to the temporal dynamics, um, which has this very non-Markovian aspect.

So now we've got this interesting picture where things like you and me that are big, where big just means that there are lots of Markov blankets that, um, constitute.

My deep generative model, um, that enable this, um, separation of timescales.

Um, you've got this picture, um, where each level is now providing the landscape for the level below with

very, very fast changes.

But of course, these are now changing the, um, the, the, the landscape is, is itself changing.

So that now you've got a non-Markovian model of a universe that we know is Markovian.

Um, so we know because we started, if you remember right at the beginning with this notion of a random dynamical system.

That is usually associated with, uh, things like a Langevin equation or, uh, you know, a Markovian process.

Um, so we're starting from, uh, a, a, a, a Markovian, um, universe with effectively no memory.

And now we've found things inside this universe that look as if they found a non-Markovian explanation, a coarse-grained, sense, uh, sensible and sense-making explanation for their sensory impressions.

That is quintessentially non-Markovian has this deep temporal structure of the kind you find in language.

The whole point is say, for example, um, you know, hierarchical Dirichlet, um, um, um, process models of natural language right through to the, uh, current focus on transformer architectures in large language models.

They're all about breaking the Markovian, um, they're all about finding non-Markovian explanations for what at the end of the day must be, uh, you know, a, uh, a Markovian process.

And you may ask, well, where does that come from?

Well, um, if we spend most of our time trying to model, um, input where that input is generated by things like us, and we are complex, deep things that generate,

non-Markovian actions that, in fact, our universe now becomes non-Markovian simply because we have other things in it that are sufficiently big to have these deep generative models.

And then you go into language and communication and, uh, you know, the itinerancy, which is, I think, quite unique to us.

Uh, you know, I don't think you'll find this kind of dynamics if you get too big.

Um, and certainly as we've already discussed, it can't exist at a very microscopic scale.

So even in your head, um, your macromolecules in the 16th, um, dendrite in the CA1 field of your left hippocampus, that doesn't do any planning.

It's too small.

Uh, it doesn't have that requisite depth of, or nesting of Markov blankets.

Um, uh, so small stuff can't do, can't have this separation of temporal scales and this particular deep itinerancy and, um, sort of, um, you know, Markovian breaking like property.

Um, but interestingly, when you get too big, um, you also can't do it either.

So, you know, the motion of the heavenly bodies, the, uh, the moon and the, and the sun and the like, they don't plan.

Um, and you know, it's interesting to ask, well, why not?

Well, because they don't have, um, because a lot of, because they're so big, all the random fluctuations in their world have gone away.

Um, and you now, um, are left with classical mechanics.

So, you know, you could sort of get deterministic chaos and n-body problems, but there wouldn't be a description, um, that would, um, be usefully interpreted in terms of planning as inference or active inference

in the way that we've been talking about.

So I think there's a Goldilocks regime.

You've got to be big enough, but not too big, uh, to do this, uh, to do this kind of planning.

Wow.

Amazing.

Yeah.

I mean, it's, um, it's bringing up all of these existential, potentially fluffy questions in my mind, um, which I, I, I would like to sort of ask at least one, which is we could ask why let's take the macro structures, the celestial bodies, black hole, the universe itself.

We could ask why, why in a mechanistic mathematical sense, do they not, um, have this kind of non-Markovian temporal horizon?

So there's a mathematical answer to that, which, or a physical, physical, in the sense of being explained by physicists as an answer to that.

There isn't necessarily an existential answer to that.

Um, and I guess I, I do not expect a concrete answer because I'm sure there isn't one, but millennia has been dedicated to the question of the human condition and why maybe that itinerancy is so fundamentally defining of our nature.

Um, which is that we can't seem to kind of sit still.

We are open and the world is affecting us.

So we're affecting the world, as you said, and we're in this constantly dynamic coupling.

I guess I would love to just hear intuitive thoughts that you might have on why the universe would unfold in such a way that you do have a Goldilocks zone.

Um, why, why, why is it the case that things like us have this wandering openness rather than just being, um, either in the soup or being firmly rigid forever?

If you have any thoughts on that.

Um, you might, yeah, not well formed thoughts.

Um, but I mean, it is a fascinating thing to reflect upon.

Um, I, I think the answers that someone like me would give you would inherit from a, uh, sort of, um, a classical view of, um, physics as opposed to a quantum information theoretic view.

Uh, so you may get a very different answer if you ask somebody like, um, um, um, Carlo Rovelli or Stephen Wolfram or, um, you know,

my friend Chris Fields, um, but if you're happy with a sort of, um, 101 physicist's response, uh, to that, um, the way that I would look at that is through the lens of the renormalization group,

which just says that there, um, that, um, that, um, that, um, that things exist at different scales.

And there is, uh, um, you are, um, looking for, uh, explanations of those that, that, that, that scale free behavior in terms of laws that are dynamics or Lagrangians technically, um, uh, that are conserved, um, over different levels.

And for, for, for the free energy principle, that is just the, um, the dynamics, the gradient flow on the, um, uh, on the free energy defined at each level or each scale.

Um, and that leads you to then a view of, um, the same kind of things, sorry, the same kind of processes and

the same principles applying at each and every scale of the universe.

So that you would add a very, very fine time scale and a very, very fine, um, um, um, spatial scale.

So in the quantum scale, you would have, uh, lots of very, very fast, hot stones moving around, uh, where the random fluctuations predominate.

Um, and there is very little of this kind of long-term itinerancy.

Um, now technically that results from, um, um, um, some, uh, a certain aspect of the way that you can always decompose any dynamics, any gradient flow.

So I've spoken, um, about the dynamics exclusively in terms of gradient flows flowing down, um, the, uh, flowing down the, uh, the, uh, free energy landscape.

Say on a Cisco manifold being like sort of favorite picture of that, um, that denies a really, another really important aspect of the flow, which is the flow orthogonal to that gradient flow, uh, known as solenoidal flow or divergence free flow.

Um, so the way that I look, that I look at this, um, through the lens of a sort of classical physicist, that, that would predicate everything on this random dynamical system or launch run approach.

Um, is that you've got a continuum between the very small and the very big, where now we're not talking so much, um, about sort of the, the degree of nesting of Markov blankets, but simply the, um, the, the contribution of random fluctuations to the dynamics.

Um, and the important thing to bear in mind is that there is, uh, any dynamics, any flow has this solenoidal part, this conservative part, and this dissipative part that inherits from the random fluctuations.

So if you're very, very small, then all the random fluctuations have not yet been averaged away.

And the dissipative part predominates over the conservative part, the solenoidal part, um, which means that at the very, very small scales, you are effectively in the world of, um, thermodynamics where you're effectively ignoring the

the, uh, the solenoidal part.

So everything is dissipative, you get fluctuation dissipation theorems, um, you know, um, uh, integral fluctuation theorems, uh, you know, and, and then you can drive generalizations in the second law and all of that good stuff.

Um, and then as we, as we get bigger and bigger and bigger, the, what the coarse graining implicit in the, um, what's known as, uh, uh, um, an RG operator in the renormalization group.

So, you know, I look at it as a grouping and dimension reductions, of course, graining, uh, or entails an averaging.

And as you average, the random fluctuations average out to zero.

So now the, the dissipative part disappears and you're just left with a solenoidal flow.

Which is just the conservative dynamics of Newtonian mechanics and Lagrangian mechanics.

And the solenoidal aspect is just a description of the, like the moon going around the earth and the earth going around the sun.

It's things that have orbits, stable orbits or quasi-periodic, um, um, orbits that are approaching deterministic simply because they are very, very big.

So the interesting thing happens in the middle where you've got a mixture of dissipative, uh, dynamics and conservative dynamics.

What would the conservative dynamics look like?

Well, they would look like this sort of, um, itinerant, um, um, um, oscillation.

It would look like the same kind of, uh, classical conservative oscillation that the moon does, but it will be much more itinerant.

It'll look like, uh, uh, uh, um, a biorhythm.

It will look like a life cycle.

It will look, it will have that property that we associate with living movement.

So, you know, we talked before about the fact to, to, um, to be autonomous is to move.

It is to have active states.

One would also argue that movement in and of itself is not sufficient.

You know, the moon moves, um, the oceans move.

Um, are they biotic in their self-organization?

I think you'd be looking now for a special kind of movement, which emphasizes the solenoid, a mixture of solenoidal and dissipative, uh, dissipative dynamics.

Um, so sandwiched in between the very big and the very small is a Goldilocks regime, where indeed you do have this opportunity for the biotic kind of self-organization that characterizes our existence.

Um, that is, um, not only, um, manifest in terms of, uh, delicately crafted solenoidal diversion three dynamics, but also nested at different timescales.

So just think about all the different timescales that your, um, body entertains solenoidal dynamics, um, um, rotational dynamics from the fast gamma oscillations, again, in that, uh, hippocampal neuron.

So you're on through to, um, your, um, cardiac cycle, through to your respiratory cycle, through to the, your diurnal cycles, through to your, um, um, slower cycles, right through to your life cycle, uh, and so on and so forth.

So then you take it up to any scale you want.

Um, but the key thing, the point being made here is that not only do you have this admixture of dissipative gradient flows that underwrite self-evidencing.

This is in the context of this solenoidal flow at multiple scales, at nested scales.

So again, we come back to this deep nesting.

Um, so on that view, there will be a Goldilocks regime.

And of course it is only at that, in that regime, will you ever get systems that start to self-model and communicate and co-construct their, their niches.

Will you get cultural niche construction?

Will you get the kind of, will you get language?

Will you, will you have conversations like this?

It's not going to happen, um, at the level of, uh, uh, macromolecules or cells, and it's not going to happen at the level of heavenly bodies, but it will happen at our level.

Um, um, uh, and clearly has happened if I can believe my senses.

Yeah, I guess the, yes, I guess the, I mean, my followup, my, my inevitable followup would be why are there these different sizes, but I don't, I don't want to get into that book.

And it's just turtles all the way down about why.

Well, no, no, no, I mean, we can, we can bring it to close very quickly, but that's it.

It is turtles all the way down.

That's I, I think a fundamental insight.

Uh, and once you, once you commit to that, everything starts to make a lot more sense.

So, um, you know, just in terms of, um, you know, how long will my Markov blanket last?

Well, it will last, you know, for a long time at my temporal scale, but the, um, the scale above, um, will last much longer.

So I have an environment and a niche that lasts longer than I do.

And my body lasts longer than anyone's cell.

And my cell lasts, lasts longer than any one intercellular component.

And, uh, we are, that, that component will last longer than any particular macromolecular configuration.

So, uh, so it is all the way down and you need it at each level.

You have to have the context being a thing, having its own Markov blanket at its scale in order to, um, provide the context for the faster coming and going of Markov blankets at the lower scale.

But in the same sense, because the larger scale inherits from the dynamics of the lower scale, there's a circular causality, which people like George Ellis would cast in terms of top down and bottom up causation.

Put simply what that means is I cannot exist.

Well, let's put it another way.

My hippocampal cell cannot exist unless my brain is in good shape.

My brain cannot be in good shape unless I'm doing good active inference and relating correctly within my family and my conspecifics.

My conspecifics cannot, uh, work, uh, exist unless there is, um, a biosphere that sustains that kind of life.

The biosphere cannot exist and so on and so forth all the way up and all the way down and every level there are things and those things have to have Markov blankets and every level that can be construed as doing some kind of elemental inference or self evidencing.

Um, the, the story we're telling there was the special kind of self evidencing that comes out at a particular scale, but the existence of that scale requires exactly the same.

Um, I repeat from the perspective of the renormalization group, it requires exactly the same dynamics or Lagrangian or functional form of the dynamics, um, at, uh, you know, at every level.

And, you know, on this view, because the explanatory target or the free energy principle is basically relational.

It's trying to understand self organization, um, of a thing that can be individuated, um, in this context and in terms of what it is context.

It's a very myopic, um, ambition.

It doesn't describe, you know, cosmology or, um, you know, the, the universe.

Uh, you're not going to get electromagnetism from the free energy principle.

Um, um, uh, and so it doesn't really have to worry about when does this, you know, when, what is the largest scale?

What is the smallest scale?

You can keep on going all the way down.

Yeah.

And if you had to hedge your bets, cause I guess it is a hedge better bet bet hedger rather.

Um, does it stop?

Does, does the, does is it, or is it just, it just goes on Markov blankets on Markov blankets?

Yes.

Yes.

Mathematically.

Yes.

Yeah.

There is.

Yeah.

It's a little bit like you asked me, uh, where's the, um, um, where's the edge of the earth?

When does the earth stop?

Yeah.

If you, if you, if you're committed to the renormalization group, um, no, it just keeps on going forever.

Um, that, um, um, and you sort of, I, I, I quite comfortable with that because, um, you know, it, it emerges from a sort of, um, a question, which is what is a state?

So we've been talking glibly about state spaces, statistical manifolds and states and fairs that we're in states and latent states, um, to have a state space and particularly, uh, specifically to have a, um, a state space that could be, that can be interpreted or construed as a statistical manifold.

You have to have states, but what, where do states come from?

Uh, and the argument, sorry, I shouldn't ask you.

I do.

The argument is that they are the states of things.

Where do things come from?

I would just said, it's a partition of state space.

Where do the states come from?

Um, if you try to answer that question, um, then you, all you have to do is basically, um, determine the mapping between, um, things or particles and states.

And the answer is quite simple.

It is, it is basically one way of looking at this sort of, um, or using the operation, apparatus of the renormalization group.

Um, the, the, the state of a thing is the, uh, state of its Markov blanket states.

Um, um, then the, um, the, the, then that state itself now has a Markovian or a, a, a partition

into Markov blankets.

So you now get back into blankets and blankets.

Um, and so that you can resolve the paradox or the, um, the, the catch 22.

Yeah.

What's the state of something?

Well, what's the thing?

Well, a thing is, is determined by, uh, conditional independence among the states.

Uh, you know, you can resolve that infinite regress just by saying it is just a recursion.

The, you know, states at one level inherit the averages of the Markov blanket states of things at the next level.

But the, at every level there are things.

So you have this, um, you know, you just keep on going down.

I should say, um, you know, there are people who believe that, um, there is a smaller scale or it is sufficient.

I'm not absolutely sure about this, but if you ever get the chance in later life, if you can talk to, uh, again, people like Carlo Rovelli or Stephen Wolfram, um, you know, from the point of view of quantum loop gravity and from the point of view of the Ruliad, um, then there, you know, there, there, there is this notion that there is some irreducible, um, um,

um, um, smallness scale, uh, beyond which you can't go.

And then the game is how do you get a continuous world out of this sort of quantum discrete, um, um, very, very small description of things.

Yes.

Yeah.

Yeah.

Yeah.

Strange loops makes me think of Hofstadter and strange loops also makes me think slightly of the, um, holographic principle, what you just mentioned there about projections onto the screen.

Um, this, this, this notion.

Um, reading off the Markov blanket, you said something that I said, we had three things I wanted to talk about.

Um, and we've covered temporality, albeit over a long temporal scale.

Once I very quickly ask, you mentioned, um, in passing, um, about celestial bodies being nearly deterministic and my ears pricked up.

Um, I, on an internalist model of the free energy principle, I sense that there's actually a kind of licensing for free will because we have this internal model and we act upon the world as to confirm the model that we, we wish to, we wish to observe.

On an externalist perspective where we really are no different from another physical entity, except for the fact that we have the possibility of moving things and, and acting to, to self evidence.

There isn't so much space for free will.

Um, but my ears pricked up when you spoke about agency.

And when you spoke about the celestial bodies being nearly deterministic, because it makes me think it seems to imply.

At least, uh, heuristically that being open to the environment, being an itinerant being licenses agents agency, but I'd want to make sure this is true metaphysical agency in the sense of not hard determinism.

Um, or whether this is agency in the sense of self conception as a free agent, because those are two different things.

Um, it's, we, we, we, we, we won't be denying that we perceive ourselves to be free.

I guess the fun, the more fundamental question there is whether that's an illusion or not.

And so I was wondering whether free energy principle has anything to say about a kind of libertarian versus deterministic perspective on free will.

Um, well, I mean, you, you could certainly use the, the, the, um, the free energy principle to tell stories that sort of, you know,

Speak to those arguments.

Um, and I think you picked up on some, you know, some of the key parts of those stories.

Um, so, you know, deterministic chaos that you get with embodied problems when thinking about some massive bodies dancing around each other.

Um, you know, there is still chaos in mix in the mix.

Um, so there's an unpredictability sensitivity to initial conditions in, in the good old fashioned way of, um, uh, non dynamical system theory of a deterministic sort.

Um, but once you get to, um, you know, things like you and me, um, you're really in the world of stochastic chaos.

So the stochastic bit inheriting from the, uh, the random fluctuations where you've got, um, two kinds of, um, if you like, or two aspects of uncertainty and implicitly then in modeling that uncertainty, um, um, uh, the opportunity to talk about sort of free will, uh, and selection.

Um, one interesting twist here is that the, um, if you watch, you want to apply the free energy principle as a principle of least action to things like you and me, uh, what you have to, um, um, um, imagine is that we are deterministic things in a stochastic and chaotic world.

Um, so in making sense of things, we only do so at a particular scale.

We're talking before about the averaging away when I'm sufficiently big to average away my random fluctuations under the free energy principle, that literally means you have to take the average of lots of neurons.

So the kind of sense making the kind of, uh, dynamics that you would simulate in the free energy principle applies to, and only to the ensemble behavior where you can average away any of the stochastic aspects.

So you will not get, you will not be able to deduce evidence or, um, um, solicit, uh, or, um, find evidence for the free energy principle by looking at single neuron or single synaptic.

It has to be at the, the ensemble level where the, the, um, well, there are all sorts of arguments where that has to be the case, but, um, from our point of view, it's when the random fluctuations are averaged away.

Um, so, um, um, I'm disqualifying what I'm saying is that I'm not saying that the brain is a stochastic or shows stochastic chaos.

What I'm saying is that the brain has to provide an apt explanation or approximation, um, to explain stochastic chaos.

Um, and crucially, um, the, um, stochastic chaos associated with our own behavior, um, with the consequences of our own actions.

So you've got a deterministic system trying to now model a non-deterministic stochastic chaotic system.

And, and you know, in so doing, well, that's easy to do because of course we're, we're talking about, um, that modeling being in terms of belief or probability distribution.

So that's, that's absolutely no problem.

Um, so, um, on that view, I think you could easily motivate, um, free will of a certain kind, um, in the sense that you, when, uh, when you're planning, you have effectively to select amongst different models of the future.

Technically, so we base a model selection and selection is an act.

And so again, we have a different kind of agency here, but it has, it's a kind of agency inherent in the selection amongst competing plans or policies, uh, narratives, paths to the, into the future.

Um, that I think has all the space to, um, um, say that this, and it's you doing the choosing.

Remember you can't, there's nothing on the outside that can influence directly that choice on the inside.

So from your point of view, you certainly have free will.

Would it look like you have free will from the outside?

Um, um, I think that's a more vexed question.

Uh, of course you'll never know, but, but, uh, I think you'd, you'd probably, you'd be in the game of, of, of, of, of a game, practically how I spend most of my, um, day job.

Um, day job is inferring to what extent is this artifact or this person, um, doing good, um, planning as inference, good planning.

Um, you know, for example, if somebody is in a coma or in the vegetable or asleep or has a psychiatric disorder, um, you know, you, you need to know to what extent they are.

Um, and in what way they are self modeling.

Um, so, you know, um, this is a slightly sort of clinical and abstract example, but it illustrates the point that you will never know whether something has free will.

All you can do is infer that they are behaving as if they had a generative model that entailed a self model and, um, uh, and thereby a model of the consequences of action.

And therefore, as if they were a true agent and possibly they may know they're a true agent.

They may not, you know, they may be, again, um, you know, that would require another level of, of, of the Markov blanket.

Is that what you had in mind?

Yes.

I mean, I think, um, I guess the, the, the kind of hardline determinist response would always be pushing you to say whether that organism or particle could have done otherwise.

So it may have this reflexive quality of, okay, I'm doing Bayesian model selection to my sense of self.

And we will come to consciousness because I think that's the kind of crux of this issue is why is any of this happening online anyway?

But it might have this self, um, a flexive notion of being a free agent, but the hardline determinist from the kind of external position of physics might always say, well, could that bundle of atoms ever have done otherwise in

terms of its Bayesian model selection?

Um, but it's, it's, it's, again, it, it might just be one of these sort of loops that just keep spinning.

Um, yeah, it's, it's all, um, fascinating.

I had a question that I've just popped out of my head when I was talking about free will, which was, um, I will come back to it.

I will, it will come back to me.

Um, consciousness.

I know it's, I don't know if it's being considered, I don't know if we can call it a red herring yet in, um, the academic world because it's become a very hot topic.

Um, and people are getting very heated about it and there's no need to bring up particular examples of when people have got heated about it, but people are very invested in their theories.

And I guess rightly so.

It seems to be this one thing that people can't seem to couch within their theories of everything.

If they have one, do you, let's starting with a very simple question.

Do you consider it to be a real problem?

Do you take something like the hard problem seriously?

Or for you, is it more of a meta question or can it all does a deflationary account just do the job in terms of the hard problem?

Um, it really depends who I'm talking to.

As my friend, Jeff Beck would say, it depends which pants I'm wearing.

So I think, I think very much it's, it's a question of, um, a conversation and you have to, you have to infer how much investment the person you're talking about has in this issue and what position they're taking.

Um, and I mean that, you know, just in terms of being polite and engaging, you know, with integrity and scientific, uh, debate or indeed philosophical debate.

But I also made it, uh, in a slightly more fundamental way that consciousness is, uh, is an illusion.

Um, and, but I think illusions are fantastic.

They're literally fantasies.

And I think that's what makes our brains fantastic organs.

I think they're hypotheses.

I think that the illusion is, um, the thing that we, uh, that we test.

It is the thing that generates the predictions in predictive processing.

So I think consciousness is another illusion, um, very much along the lines that one reading of Andy Clark's base in Qualia paper, um, um, the even qualitative experience Qualia are just illusions.

They're just things, they're just constructs we bring to the table, including selfhood.

Um, they're just hypothesis illusions that are particularly apt at, um, explaining everything that we sense and making good sense of it as, as a, a parsimonious hypothesis.

Consciousness itself, I think is exactly that because it only applies.

Consciousness, the hypothesis is that, um, is this thing conscious or not only applies because there are other things in my universe that I'm trying to have to model like you.

So it's, the question is not whether I am conscious.

The question is whether you are conscious.

Consciousness, and of course we've already said that is unanswerable because you have a Markov blanket.

So it's an unanswerable question, but crucially, it's a question about the things I observe that I'm trying to make sense of.

Um, it only becomes, I think, um, only in, in the spirit of the meta hard problem or the meta problem.

Um, it only becomes, um, um, vexed when you make the mistake of asking, am I conscious?

Uh, as I don't think that's what the question is there for.

That's not, the illusion is not for me.

It's for you.

Uh, the illusion is the consciousness is a construct, a hypothesis and a very plausible one.

All the evidence, uh, that I have at hand would suggest that you are conscious.

And indeed, literally, because there is that evidence, um, it is there in the sense of an evidence maximizing illusion hypothesis or component of my generative model.

Um, so I, I go through that just to make it clear that the, you know, the answer depends upon who you're talking to, but also who you're trying to, um, ascribe consciousness to.

So if I was talking to, um, um, well, if I was talking to, um, um,

people, uh, like Jacob, how he, um, um, you know, the first thing he'd say is that the free energy principle is not a theory of consciousness.

Uh, and in fact, he likes saying that every, every, every opportunity.

So, so there's nothing, there's nothing, um, there's nothing in the free energy principle that has anything explicit to say about, explicit to say about consciousness.

On the other hand, people have used it in different ways to, uh, address the hard problem.

Um, uh, you could take the view that Andy Clark takes that there isn't a hard problem.

And perhaps even with David Chalmers' work recently on the meta problem, the meta hard problem, that's probably a more interesting problem.

Um, or you could, you could grasp the nettle and start to talk about it.

If you were talking to, um, my colleagues, uh, Chris Frith and, and, um, um, um, and, um, um, Adam Safran and Maxwell Ramstad and, and, and, uh, uh, whole, uh, this, and that, um, certainly, um, so I think inspired, uh, by something that you mentioned earlier on, which was the holographic principle in, um, um, in, in, in quantum

information theory where the holographic screen now, um, plays a role of, of the Markov blanket that separates the inside and the outside in terms of bulk reading and writing to, um, the, um, the, um, the, um, the holographic screen.

Um, if you're talking to him, um, then he would start to talk about these minimal screens, uh, these minimal Markov blankets, um, and develop quite a, quite a sort of nice argument that if you can't reduce in the sense that we're talking about nested Markov blankets inside a Markov blanket, if there are no more reductions at hand, then the only way that the internal states of that irreducible Markov blanket

can know about their own existences by acting upon lower levels.

And therefore there's some, um, there's something unique, there, there exists a unique Markov blanket that,

that, that, you know, um, maybe, um, uh, have the attribute of, of consciousness.

Um, if you were talking to somebody like, um, Vanya Weiss, um, um, he would, uh, you know, tackle this in terms of, um, trying to resolve, uh, a duality.

And, uh, dualistic position by offering a dual aspect monism, um, uh, of a Markovian sort, and that rests very heavily upon this information geometry.

So if you remember before, we're talking about sort of a state space equipped with free energy functionals, um, on it, um, um, that, um, plays a role of a Cisco manifold.

So if you remember, you know, a Cisco manifold, but, but there are two Cisco manifolds that, uh, jointly sit up, sit in this, in the same state space of, say, neural activity.

One is that for any given, uh, point, there is a probability of being there from a point of view of neuronal activity and the thermodynamics.

So this would be a thermodynamic, um, kind of information geometry.

And then the other one is what the free energy principle uses, which is the encode using that point to encode beliefs about external states.

So you've got two, um, information geometries, belief structures, if you like, um, that supervene on exactly the same, um, material, um, process.

One of which is pertains to probability distributions or beliefs about the internal states.

And the other one is probability distributions or beliefs encoded by the internal states about the external states.

So you could construe one as being the mental and one as being the material.

And then you've got this dual aspect monism where you're now asking questions about, well, what are the lawful relationships between the thermodynamic free energy and the energy?

And the, um, the, um, the variational free energy that pertains to beliefs about things.

So you've got now a mechanics that talks about beliefs and that may well be one way of repairing that sort of a dualistic approach.

Um, there's another, uh, another perspective, which you would get if you were talking to, talking to people like, um, uh, Giulio Tononi and, um, Cyril, um, uh, well,

my colleagues involved in, uh, one of the Templeton Foundation, uh, abseiro research, uh, um, collaborations, um, where you're trying to find bright lines between the free energy principle and integrated information theory.

Um, um, um, um, which is not necessarily the best thing to do because a lot of integrated information theory is exactly consilient with the free energy principle.

However, if you are going to get funded, uh, by, uh, trying to tease them apart, then the thing that distinguishes those approaches to, um, uh, consciousness is really the, uh, the active part that we're talking about.

So, uh, boiling this down in a way that somebody like, I repeat, Jakob Howie might, might articulate this is, it's the difference between seeing and looking.

You know, to be conscious, I have to look, it's not enough to see, um, it's, um, you know, to be conscious, I have to listen, it's not enough to hear.

Um, so you're putting action into games as active sensing, this sort of closing the loop between the inside and the outside.

Maybe, um, the thing that, um, is necessary for consciousness.

Consciousness and simply because you're saying that consciousness is an attribute of agents and agents have to act.

What do they act on? Their sensations.

What does that mean? They do active sensing.

So to be an agent is to be an active sensor.

To active influence.

Um, and you know, if you're saying that things that aren't agents can be conscious, then you wouldn't worry about that.

But if you are committed to a deep, um, to the notion that to be conscious, um, of, to have a minimal selfhood, um, then you have to have the notion of self.

If you have to have the notion of self, then you have to be an agent.

If you have to be an agent, you have to, um, um, have generative models about the consequences of your agency, your actions, uh, and that has to have the temper.

And then we go through the, all of those nice links that we were talking about before with temporality and, um, you know, and the like, and indeed that hierarchical structure takes us back to the Chris Fields, um, irreducible, um, uh, holographic screen or, or, um, or Markov blanket.

So that would be the way that I would, um, converse with that crowd.

Uh, uh, I've run out.

Yes. I've run out of people that I might talk about it.

Uh, so I have to find out what you like and then I'd, I'd tell a story that you like, or let you tell a story that, uh, that you like.

Well, I guess my question to you would be, can you imagine a world of self evidencing?

Can you imagine, okay, let's put it another way.

Can you imagine a perfect replica world like ours where the lights are turned off entirely?

There is no consciousness or is it, do you imagine it's a necessary, uh, consequence of the way that creatures are like, uh, like us are structured that to do deep temporal planning, to do planning as inference, whatever it is.

Or self hooding.

We need consciousness.

Could there have been another way that the universe would have unfolded without us being online?

Um, I suspect not.

Um, sorry.

I, I, I'm torn between, um, um, being taken by you down sort of, you know, philosophical zombie, uh, roots versus, um, another, um, story that comes out in a free energy principle about consciousness, which is, um, the, um, the, um, the, the, the importance of self modeling in relation to other modeling.

And the fact that we are actually living in a world, um, of things like us.

So, you know, one story I didn't cover, which is a story you would tell if you were talking to, um, my friend Chris Frith or people who are much more interested in the sort of social neuroscience and, uh, or say

attachment theory, um, and, you know, neurodevelopmental issues, um, that are induced by a sort of evolutionary psychology view or niche construction, for example.

All of these views of consciousness means that consciousness is just there, um, which is where I started to infer, um, the state, the disposition, the intentions, state of mind of somebody else.

Um, why and how can you do that?

Well, you can only do that if that thing is sufficiently like you to be able to use your generative model, um, as, uh, obviously your physical actions as a model of their physical actions to infer their intentions and the like.

Um, and when one takes that to its limit, then, um, often what the real problem is, is not in terms of inferring the content of some communication or some, uh, uh, exchange of, of, um, semiotic cues.

It really is inference about agency again.

It's did you do that or did I do that?

So that ambiguity now requires you to have a model of self versus other.

So it's a much more relational, um, um, notion of self.

So, you know, sometimes I think people talk about minimal selfhood, um, as something which is monolithic.

I don't think that's necessary as part of a gentle model.

It certainly would be important to have, um, different, um, different kinds of self.

You know, am I embarrassed?

Am I in love?

You know, as we were talking about, you know, in the, um, TMB meeting a few days ago, um, that would be, you know, an expressive, very necessary part of generative model where I have to contextualize my own emotional presence.

And all autonomic and moteric behaviors.

Um, but there's no one self is his self, you know, self like this, like this and like this, a more fundamental relationship, uh, relational aspect of selfhood is in relation to otherhood.

You know, um, you know, versus me and am I mom or is, you know, is, is, is mother thing? Am I a thing?

I'm the same thing as mom.

You know, you go to sort of Kleinian, um, uh, um, sort of, um, uh, arguments there.

So put simply what that means is it may well be only in universes or worlds that are populated by ensembles of conspecifics that are sufficiently similar.

Would you need to selfhood simply to distinguish itself from other in a universe where there's only me, I wouldn't need selfhood.

So it would not be, it would not pay its way in terms of model complexity or model evidence to have the notion of me being a self.

If there's only me, it only pays its way.

If I now have to actually entertain the hypothesis that either I cause that or you cause that.

So the question now is, is it inevitable that you get a universe with lots of similar things emerging all disambiguating themselves or attributing agency to themselves in the early loss, in the early loss zone.

So I think it probably is.

I think that would be, I think if you simulated.

Sufficiently accurately artificial life, I think you would, you would get the emergence of, um, populations of similar things and they would niche construct.

And they would bet thereby drive their own, um, uh, less increasingly nested and, um, structurally complicated, um, or sparsely structured, um, uh, organizations in terms of their Markov blankets.

And that this would, um, this at some point would inevitably invoke, um, modeling of others and therefore almost by default, um, a, a, a, a, a, a minimal notion of, uh, of, of self of selfhood.

I think that would be emergent, you know, um, but it would require being in the Goldilocks regime for a sufficient amount of time for this kind of niche construction.

Uh, cultural niche construction to actually play out.

So, you know, you'd have to have this very slow timescale providing the right kind of context for this, this kind of thing to, uh, thing to emerge.

Yeah.

Yeah.

I think my, well, I have some, I think I have some sympathies to the, to the argument that Chalmers will laid out in, or did lay out in 95 in his original hard problem paper, which is that.

I think I can still ask the further question, which is, can systems like me make the functional distinction between myself and others without there being a sense of what it's like to be me.

Can that just be an offline-ness, which is done purely in some algorithmic manner, which need not involve some quality of myself-ness.

Can that just be honest?

So, yeah.

There, there's a point that there's an element in me where I don't want to say that that, in a sense, resolves a hard problem.

Because at least I can hypothetically envision that being done offline.

Um, I wonder whether you've given much thought to Mark Soames's idea, which is that consciousness might be rooted in affect.

Because this seems to me to be a bit more of a self-confirming idea, which is that feelings must be felt.

Now, I think there's still an open question there, which is affect arises at the point of uncertainty and drives us to resolve uncertainty.

And when we do so, we feel good about ourselves.

Now, again, I still think when I was reading his book, I still had this open question.

Well, why would that have to, why would, why would affect even emerge in that place?

Why wouldn't we just have this drive to resolve that uncertainty without there being the online-ness?

So, at least for me, although I may be wrong, and I'm very happy to be wrong, there is still no simple parsimonious explanation which can get round this, um, this point that Chalmers makes, which is you can give me the function, but you still, you can't give me the explanatory power why it needs to be online.

Um, and I wonder if there ever will be some mechanism which is sufficiently, which isn't just purely tautological or invokes consciousness as some fundamental reality, which makes me, which makes me wonder whether you've heard of Donald Hoffman, whether you're familiar with his work.

I certainly have.

Yes.

Because I sense that in some ways it has some interesting implications maybe for the free energy principle in the sense of space and time not being fundamental units of reality.

Um, we've been speaking a little bit about how, um, where do space and time come from? How much do space and time overlap in the sense of generating, um, like we need expansive, expansive creatures like us with embedded Markov blankets inevitably have a sense of temporal modeling.

And there's these interesting interactions. Does the idea of space and time being what he would say applications on a desktop. Does that have any real, do you sense has any real ontological implications for free energy principle or is it a red herring?

Um, well, you've, you've, you've introduced two important things. I forgot to mention Mark Soames is another friend.

Yeah.

So if I'm sorry to Mark Soames, I would say it's all about this irreducible Markov blanket that acts upon the rest of the brain. Where is that action? It's going to be in the neuromodulator systems that set the precision.

The precision is exactly that Fischer metric, which tells you about how much you're updating your beliefs. It is the way that you do mental action.

It is as you intimated, um, feeling in the sense that it is quintessentially balanced by increasing or decreasing the precision or uncertainty.

So, um, you know, that's how I would tie in Mark's, um, formulation to, which I think is internally consistent with, um, say Chris Fields formulation or, um, uh, Vanya's formulation.

So Mark so focuses, you know, on us, I think a key aspect of, um, of active inference, which is, uh, if you're a psychologist, it will be attention.

It's basically acting in a way on the inside to, um, get the, um, estimate and deploy the right confidence or the right precision.

And, um, um, mathematically the curvature of the, uh, of, of, of, of the, um, the free energy in and of itself, um, getting that right.

Um, and, um, you know, that has a lovely, you know, connection to attention.

It means that you can't divorce, um, consciousness from attention.

You can't divorce it from agency in the sense of mental action or internal action.

You now have a deep connection to, um, Thomas Metzinger's, um, distinction between, uh, you know, um, opaque and phenomenal transparency.

Um, and, uh, and the way that Mark would tell the story would be that this is just feeling and feeling is at the heart of, of, of any, uh, qualitative experience.

Um, so, yeah, thank you for the, for reminding me.

He was one person I forgot to cover.

Um, the, uh, the second thing, um, you know, the, the space and time being illusions.

Um, I mean, yeah, I sort of start off there.

So I, I, I, I'm not quite sure that it, uh, so if you're saying that those, um, and I should say, he's a friend, I think I've never met him,

but I think he's a friend of Chris Field and Mike Levin.

So, yeah, there's, there's a, a small world here.

Um, but if you're just saying that, um, space and time are, um, illusions in the sense that we were talking about illusions as being fantastic.

And the thing that makes us into fantastic little beings and have beings and, um, uh, and our brain into a fantastic organ.

Yeah, absolutely.

I mean, you know, the question for me is why on earth would you, um, what, what, what must be the case for us to have a world of science?

It's for us to have this illusion of space time and that we are living in a metric space and it's practically very important.

I mean, a lot of people spend a lot of time in machine learning, embedding things in metric spaces.

Um, and sometimes it's very difficult to see what licenses is there to do that.

Uh, you can certainly see how it would work in terms of projective geometries in metric spaces.

but the deep question is why you know you know I can't I can't imagine a you know an amoeba really having having a notion of space I may be wrong and so knows you know if you listen to people like Stephen Wolfram you know the notion of space-time itself may the illusion of space-time itself may be just an artifact of the fact that our sensory organs respond very very slowly in relation to the speed of light so if light traveled very very slowly or imagine a world in which we saw with sound and sound was very very very slow would that would that be would that give you the illusion of space you know a distance as an action of the kind that you know we sort of implicitly assume when we're talking about sort of visual objects so I think he has this nice notion that you know in this kind of world with these kinds of sense organs you know when something moves from here to here is it the same thing anymore because that's what you're saying when you have a metric space in which objects move you know some certain symmetries or invariances inherit from the notion of movement in a metric space but that may well just be be be a fantasy we've got gone off quality of experience and the hard problem yeah well no I mean by the way it all ties back it's a beautiful it's it's a very good question which is why would this why would space and time be a in the interface for us I guess that is the deep question um and yet it appears that at least it is something at least in conscious awareness I did remember my question and it actually feeds into the last couple of questions I will have which is you mentioned um that you spend a lot of your time inferring in the technical and layman sense whether other people have I think you say good generative models there's an interesting question here I think which which I may have had intuitively when I started reading about active inference which is is there any such thing as good self-evidencing in the sense that I may so so a lot of as of course do you know as you wrote a lot of papers a lot of psychiatric disorders and psychological disorders have been rooted in um disruptions to precision weighting mechanisms for example can we consider are the words good bad optimal suboptimal are these useful terms in this context or is it that that individual has a set of priors which are different from you and I and everyone else and we all actually have a sense of we all have kind of different priors so all self-evidencing in unique ways

and yet philosophical vagueness allows us to clump us together into a group
is there such a thing as what it is to be a good self-evidencer as a human
with beyond beyond just a trivial point that we're not just dissipating into the the heat bath
right yeah that's an excellent question um so I think the answer to that is yes um but as you
uh correctly um pointed out they should be qualified by the um the complete class theorem so you you're
clearly referring to the complete class so they might be just wise to um just define what that
means and the implications for discussions of good and bad um generative models um so the complete class
theorem the way I like to summarize it is that for any given pair of behaviors and loss functions there
exists some priors that render your behavior pays optimal which means that there is always a
description of every behavior as base optimal under some prize so in that sense you can't be good or bad
um where the thing is basically your decisions because this inherits from basic decision theory so
it's not quite as full as uh active inference in accommodating um information game but um within
the within the um confines of basic decision theory you can't be um you can't be good or bad you just have
different priors so you know my favorite example is that you know if we were all went to live on um on
mars or a much heavier planet people with parkinson's disease would probably be much better have better
prize you know uh sub personal prize in terms of topia deficits because you know they move more
slowly and then sort of more cautiously yeah um so yeah I think from the point of view the complete
cluster I think it's important to bear that in mind you know there are just different prize but I
I do think though you can sort of just say well hang on a second what about whether a good and bad prize
and I think then we get into this again this hierarchical story because of course as soon as you have a
hierarchical generative model there are no prize you know everything's an empirical prime everything
can be contextualized everything is conditioned before everything else including your prize and then
you're just back into the um into the game of self-evidencing what is good self-evidencing it's
achieving a higher marginal likelihood or model evidence um how would you do that by finding the right
prize how would you do that by learning how do you do that by changing your model with basic model
selection or structural learning um how would you do that at an evolutionary scale you do basic
model selection with natural selection so yeah um on that view the log evidence as bounded by the free
energy um is just adaptive fitness is how well your priors are adapted to this world you are trying to
explain so I think yes absolutely you can you can you can be a good self-evidencer you can be a bad
self-evidencer if you're a natural selection um that matters because if you're a bad self-evidencer that
means you won't be around um in the um in the next generation so the marginal likelihood of you existing
at an evolutionary time scale is low so the your your evidence is also the probability that you will be found
exist you will exist so it's quite important to get to be a good self-evidencer and because that's your
marginal likelihood of existing uh and I've now slipped in another synonym for um free energy beyond um
prediction weighted prediction errors or beyond surprise or surprise or self-information beyond log evidence
it's also it's just it's called the log marginal likelihood it's the the likelihood of your
sensory exchanges with your world your eco niche having marginalized out or averaged away all the
unknowns which in this context are all the states out there you know the the hidden causes or the sorry

the external states which you model in terms of hidden uh in terms of hidden causes um so yeah I think it's very important to be good at bad at evidencing the measure of goodness and badness is just the marginal likelihood of the model evidence and the free energy there is nothing else um uh if you're saying does that speak to some creatures that will be able to actually assess how good they are at self-evidencing then yeah I would say absolutely um people like uh Matthias Joffoli uh looked at this specifically from the point of view of emotion and developed a whole taxonomy of emotion in terms of um rates of change of free energy and you go into stories of slope chasing uh um looking for gradients and gradients and gradients of uh self-evidencing on the measurement of the goodness of self-evidencing which is uh which is uh which is which is the free energy um more refined stories more recent stories um would basically be looking at certain kinds of uncertainty um being encoded explicitly in your generative model as underlying affect uh particularly the the your certainty about what you're going to do next for example then we get back to um the story that mark soames would like to tell which is um the precision of your beliefs is the most you know the most important attribute that defines um you know its valence and that you're you're feeling at this sort of minimal effective level yeah yeah the complete class theorem is an interesting one because I think it leads people it leads one into a potent these potential strange conclusions um it makes me think of animals that commit suicide oddly so um I think sort of aphids might explode themselves in the context of a predator it ultimately comes back down to your model evidence I guess so I am the type of creature that will commit suicide if I'm in the presence of a ladybug of course but then those kinds of examples I sense for me show that the words good or bad I'm just pushing a little bit back just intuitively the words good or back in in in the form of uh continued existence slightly put the cart before the horse because who said that that's the imperative so if I existed in the world and my everyone's imperative was to no longer exist you know god forbid then that then I would be doing bad self-evidencing on that standard if I didn't if I continued to persist so is there a point at which even the yeah like do you see how well do you see that argument holding any water um or is it just now we're just sorry dissolving into semantics uh well no no I think I think it's an interesting thing to to contemplate uh you're you're you're basically I think undermining um that particular argument by uh um a proof uh abductor absurdum um so if you think you are the kind of thing that will uh self-destruct or to vitiate or commit suicide um then uh you can't exist because you've killed yourself right yeah so I will be extremely surprised um if there were any um uh common natural um um subhuman artifacts that uh kill themselves I didn't know about the apid example I'm very I mean obviously lemmings is is the sort of popular culture one but um um you know I think they were extremely rare and you know and then you have to understand the um selection at an evolutionary or transgenerational level um um in order to apply the free energy principle um but in you know I think you're absolutely right I'm very mindful that you know a couple of weeks ago I was asked to review a paper applying active infants to suicide and it was a very artful paper um and uh I do remember thinking I'm not sure whether the point was made implicitly or explicitly in the paper but you know it's it made me think in the exactly the way that you were you were um talking about um you know the imperatives to survive and and are they the right way to think

about the causes of our behavior um I would say absolutely not not this is not about um wanting to survive this is simply wanting to avoid uncharacteristic sensations and then we put a postdoc hypothesis oh that's because I want to survive then right uh that's a fantasy that's another illusion uh so you know we do not know what it is like to be dead so yes it cannot be but it can be surprising but it certainly can't be the kind of surprise that you'd register in terms of a free energy so it's not that we're frightened of being dead um we are going to avoid via these pragmatic constraints um states that we've have inherited that um for portend um cessation and dissipation and death definitely in all itself is not at all frightening we go to sleep every every um suspension of course we're going to sleep every night um so you know I I would I would um sorry yeah and the final point I think is I think what you were doing though you were probably um creating a slightly vexed and self-defeating argument by saying uh by assuming that um we can understand things in terms of imperatives the whole point of the free energy principle is that there aren't any imperatives this is where it starts things that exist must behave like this and part of that behaving like this means that you can interpret them as optimizing something and complying with imperatives but that's just an interpretation that you'll bring to the table when you apply to something else if you then apply it to yourself then you know then things might get interesting you might have hard problems but there are no imperatives you know in fact even minimizing free energy is not an imperative you can have the free energy principle applying to um things that have very um dissipated uh dissipated detracting sets um that have very very um high entropy um all it's saying is that there has to be a certain uh gradient flow that counters the random fluctuations then you ask yourself well hang on a second so why does it look as if we're always trying to attain some very precise homeostasis um and the free energy principle would turn that on its head and say well look okay in some universes there may be very precise things

um and these precise things um and these precise things what would they look like oh it would look as if they were trying that they had um aspirations to evincing a very precise homeostasis and indeed allostasis so you get this notion of another kind of natural thing which are precise things and so if precise things exist it would look as if they're um they are compelled to comply to very specific imperatives but for the free energy it's not saying they exist because they comply with the imperative it's saying existence implies that it looks as if they have this imperative right now and one final point you know i'm aware that we we're now approaching three hours uh well final point though i you introduce the word optimization i think that was uh uh um the right uh a right and important um um thing to introduce because you know it's only when you bring free energy into the game as a bound on marginal likelihood or self the negative log marginalized with the self-information or surprise or surprisal do you convert an existential description into an optimization problem so that's exactly the move that richard feyman made when he introduced variation free energy he had this impossible marginalization problem that to actually evaluate the marginal likelihood of paths it's exactly the same kind of problem that we're talking about in terms of uh what are my priors over policies and plans and paths into the future um you can't you can't evaluate that exhaustively it's analytically and practically physically not realizable so the

marginalization problem the true inference problem if you like uh the true existential uh problem if there is one um cannot be solved but what you can do is introduce this bound and turn it into an optimization problem so just by using free energy um or just by appealing to this uh teleology that comes from um predicating anything on this evidence lower bound you are actually saying that it looks as if creatures that exist are optimizing something what does it look as what are they optimizing it looks as if they're optimizing a bound on their um marginal likelihood or their adaptive fitness um what does that look like well it looks exactly like bayesian inference uh or if you talk to somebody machine learning it would look as if you're um optimizing your negative variation free energy risk known as an elbow so this is a evidence lower bound it is exactly objective function used in um so high-end neural networks like variation autoencoders um so you know i think that the distinction between um just being and optimizing is quite crucial and you're absolutely right that once you make the move um of um framing things in terms of variational bounds using variational calculus or variational bays um then you have actually committed to an optimization narrative you didn't have to just to exist you you you could uh yes yeah i was gonna have this funky philosophical argument about needing to exist to predict but i won't go there i won't go there um because we are going to wrap up all of this philosophical mathematical thoughts as and consciousness thoughts has brought to my mind thomas nagel um thomas nagel has this idea that death is bad because life is

inherently good and we've been talking about life and being and what it is to be and what one needs to do to be and persist to be if we're going to wrap everything up is there a sense in which that kind of related to what we were just talking about in terms of imperatives of course there is no imperative to be in any sense but you said it was an interesting thing you said there was it looks like we are all being anyway right even if we didn't have to it looks like we are there's no god granted plan for us to be but we seem to be doing it anyway does any of nagel's romantic i philosophical idea that life is fundamentally good being is good does any of that resonate with you um yes it does because it's it's it's you know it's nice isn't it and i think you'd you'd probably you'd probably sort of license that with um phrases of the kind that you know to exist is to be curious um so this speaks to this sort of um you know the ability to respond to um epistemic affordances and certainly you know uh to exist as curious creatures um is obviously the good way to exist in your in any given world and it's you know and would be mandated um i i can't resist saying though of course this only really applies to you it doesn't apply to me so so you are good because you are curious and you exist

i wish i could say the same for me but i can't strictly speaking but you could if you wanted to i could i could abstract to you it's up to great i could infer that there is a you great um and then i guess we can wrap it up there with one more question we've just spoken about life being good i'll flip the question i don't know if you have ever heard you talk about death although we briefly touched upon it um i won't make it too personal but if life is kind of good in some sense because of the concomitant parts of living as we do as i don't know if it's good to be a stone but it's nice to be me having these conversations would that make death bad is that is there is there a is you said we fear dying because we fear in an affective sense um that which we don't expect to the states we don't expect

to embody and in some sense we don't expect to embody death because it doesn't fit in with our priors but beyond that free energy principle perspective is there a broader position in which just returning back to the fabric of the universe is bad is there something to say about re-entering that markov blanket does do my markov blankets dissipate um what what does that process look like and and does that strike fear in you or strike or um well more or i don't want to overstate it um

it is it is beautiful um from a from a mathematical point of view um so i i have a very sanguine view of this and we've already said you know you've never been dead i've never been dead so no one can tell me it's an unpleasant experience um so i i'm not particularly i i'm not going to um avoid being dead i certainly think all the all the paths to being dead are probably very surprising and unpleasant so i'm going to avoid those uh death in and of itself i don't think it's uh it's sort of um quintessentially bad um in the same way that um natal being curious and being engaged with the world is a good is it is good while you exist um um um i think that's perfectly okay but i think death is you know um the life cycle that's quite a life cycle that it's slightly less uh um um imbued with um morbidity um the life cycle is quite important because um if you just look at um if you just take this um multi-scale um um view of um self-organization cast in terms of this basic mechanics um that you know for example natural selection being basic model selection um then uh if you put that together with what we're talking about before about um ecosystems and niche construction constantly changing the lived world then what you need to do is to do constant basier model selection um in order to get the right structures very much in the spirit that we talked about before where there are good priors and there are bad priors in the sense of the marginal likelihood or adaptive fitness

and the right ones to learn those good priors you sometimes have to start from scratch again with a different structure and of course this is exactly what basic model selection does via structure learning at an evolutionary time scale will entail death so that picture basically says that we have to keep refreshing so sometimes casting terms of base optimal forgetting so we the fact that we die our base is is could be construed as a base optimal forgetting where we are the environment's memory

of what is good for this eco-dish but the environment on this natural selection knows that the environment is changing

so what was good last generation will not necessarily be good for the subsequent generation so it has to forget in the same way that

you learn by forgetting you have insights by dispensing with redundant information so in a sense death for in that life cycle

is an essential part of um uh of the self-evidencing of the species so it may be bad for you i suppose you know in a natal-esque sense in that it's denying the opportunity to be curious and engaged with your lived world

but it's certainly good for your children and your children's children

right right right i'm a i'm a forewarning for them that's good i'm glad to i'm glad to be that pawn in that game and a kind of cheery note on which to end um albeit slightly more morbid question

colt this was an absolute pleasure um despite the occasional wi-fi hiccup i think it's um it's gone

you know i've been i'm incredibly happy with it thank you so much for being our first guest
um and answering my suite of sometimes bizarre questions it was a real joy on my part right well
it's been a real pleasure talking to you your questions were excellent i have no idea how long
you're going to have to spend editing all this but i i thank you again for that opportunity to uh
you know to talk about these wonderful issues and hopefully we'll do it again excellent thank you